An investigation of age differences in the Ornstein-Uhlenbeck model of decision making

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Abstract

The diffusion decision model (DDM) has become a standard framework for studying age differences in cognitive processing. Within this framework, age-related slowing of response times has been attributed primarily to increased decision caution (wider boundary separation) and slower non-decisional processes. Here, we investigate whether the Ornstein-Uhlenbeck model (OUM) — which extends the DDM with a self-excitation parameter, k, capturing the degree to which accumulated evidence accelerates further accumulation — reveals additional dimensions of cognitive aging. We applied both the DDM and OUM to two data sets: a battery of 18 binary decision tasks ($N = 125$)and a large-scale online Implicit Association Test ($N = 5,564,947$), using simulation-based inference. Across both studies, k exhibited a robust negative association with age. In the large-scale data, k peaked in late adolescence and declined quasi-linearly throughout adulthood. Formal model comparison suggested that the OUM provided a slightly better account of fast binary decision tasks (mean RTs < 1 s), while the DDM was preferred for slower, more deliberative tasks. Importantly, accounting for self-excitation attenuated the age effects on boundary separation and non-decision time typically found with the DDM, suggesting that some of what has been attributed to increased decision caution or non-decisional slowing may instead reflect an age-related reduction in the internal momentum of evidence accumulation. Reduced self-excitation may represent a previously overlooked component of cognitive aging.

Keywords: cognitive aging, Ornstein-Uhlenbeck model, self-excitation, diffusion decision model, simulation-based inference

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In recent years, the use of evidence accumulation models such as the diffusion decision model (DDM; Evans and Wagenmakers, 2020; Ratcliff, 1978; Ratcliff et al., 2016) to study age differences in cognition has become more and more popular (Theisen et al., 2021). These models utilize both the response time (RT) distributions and accuracy data observed in simple decision tasks, for example a recognition memory task, lexical decision task, or perceptual task. Rather than analyze RTs and accuracy separately, researchers can use them jointly in order to separate a set of cognitive parameters that are assumed to represent specific aspects of the processes underlying people's decision making. In its most basic form, the DDM includes three core cognitive parameters. The *drift rate* reflects the speed and efficiency of information extraction from a given stimulus. The *boundary separation* quantifies the amount of evidence accumulated before committing to a decision, and is typically interpreted as a measure of decision caution. Finally, the *non-decision time* captures processes not directly related to the decision itself, such as perceptual encoding and motor response execution.

Over the past two decades, a robust pattern of age differences in DDM parameters has emerged (Ratcliff et al., 2001; Ratcliff, 2008; Ratcliff & McKoon, 2015; Spaniol et al., 2008; Thapar et al., 2003; Theisen et al., 2021; von Krause et al., 2022). Age differences in boundary separations and non-decision times are generally assumed to be the main model-based explanations for the well-established age-related slowing of mean response times (Salthouse, 1996, 2010): Older adults tend to set wider decision boundaries, indicating more cautious responding, and show slower non-decision times, suggesting age-related changes in basic encoding and motor processes. In contrast, drift rate differences are less consistent, appearing to depend heavily on the type of experimental task (Theisen et al., 2021), and possibly emerging in much later parts of the human lifespan than has typically been assumed in studies focusing on processing speed

(von Krause et al., 2022).

Although the DDM findings have converged to a relatively robust pattern, we know little about the age effects found when different evidence accumulation models are used. The standard DDM is by far not the only such model proposed in the literature. Some alternative models, such as the leaky competing accumulator model (LCA; Usher & McClelland, 2001) incorporate additional assumptions about the processes underlying decision making. The LCA, for instance, assumes that already accumulated evidence decays over time and that response options can mutually inhibit each other — assumptions that give rise to additional parameters with potentially meaningful individual differences. However, difficulties in recovering the full set of LCA parameters in typical experimental data (Miletić et al., 2017) have limited its use in individual differences research. As a consequence, age patterns in the parameters unique to such alternative models remain largely unexplored.

However, very recently, there has been a newfound interest in the leakage aspect of the LCA, although in a slightly modified form (Wang & Donkin, 2024; Wang et al., 2024). Leakage of evidence is incorporated in the LCA by changing the assumed process underlying evidence accumulation from a Wiener process to an Ornstein-Uhlenbeck process (Busemeyer & Townsend, 1993). The standard Wiener process includes just two parameters: a drift rate reflecting the systematic trend in evidence accumulation towards one of the two boundaries, and a random noise component reflecting unsystematic variance in evidence accumulation. The corresponding stochastic differential equation is

$$dX_t = \nu dt + \sigma dW_t \tag{1}$$

where X_t denotes the accumulated evidence at time t, ν is the drift rate, σ is the diffusion coefficient (noise), and W_t is a standard Wiener process. However, an Ornstein-Uhlenbeck process includes an additional parameter that can assume the role of information leakage in the LCA. This parameter, k, is added to the systematic drift rate and is weighted on the

already accumulated evidence. The corresponding stochastic differential equation is

$$dX_t = (\nu + kX_t)dt + \sigma dW_t \tag{2}$$

When $k < 0$, accumulated evidence is pulled back toward the starting point, representing leakage or decay. In the LCA, k was originally conceptualized as such a leakage parameter: accumulated evidence decays over time, with the pull back toward the starting point growing stronger as evidence increases. However, Usher and McClelland (2001) already acknowledged that k can also take on the opposite role. When $k > 0$, the parameter creates a *self-excitatory* dynamic: as evidence accumulates toward a boundary, the rate of accumulation accelerates, creating a feedback loop that speeds up the decision process. Importantly, recent work has shown that the k parameter is easier to recover when it takes on positive (self-excitatory) rather than negative (leakage) values, and that including self-excitation can improve model fit to empirical data (Wang & Donkin, 2024; Wang et al., 2024).

Interestingly, using the Ornstein-Uhlenbeck model (OUM) can yield novel and unexpected interpretations of well-established psychological phenomena (Wang et al., 2024). Speed-accuracy (S-A) tradeoffs — the phenomenon that in simple decision making tasks, fast responses tend to be less accurate — have traditionally been explained within the DDM framework primarily through differences in boundary separation: when speed is emphasized, people lower their decision thresholds, accumulating less evidence before committing to a response, which in turn leads to faster but less accurate responding (Evans & Brown, 2017; Lerche & Voss, 2017; Voss et al., 2004). Wang et al. (2024) challenged this account by fitting both a standard DDM and an OUM to four data sets involving experimental S-A manipulations. Model comparisons favored the OUM in the majority of data sets. Crucially, the typical difference in boundary separation estimates between speed and accuracy conditions largely vanished when the OUM was used. Instead, S-A tradeoffs mapped onto the self-excitation parameter: participants in the speed condition showed higher k values than in the accuracy condition. This finding suggests that fast responding

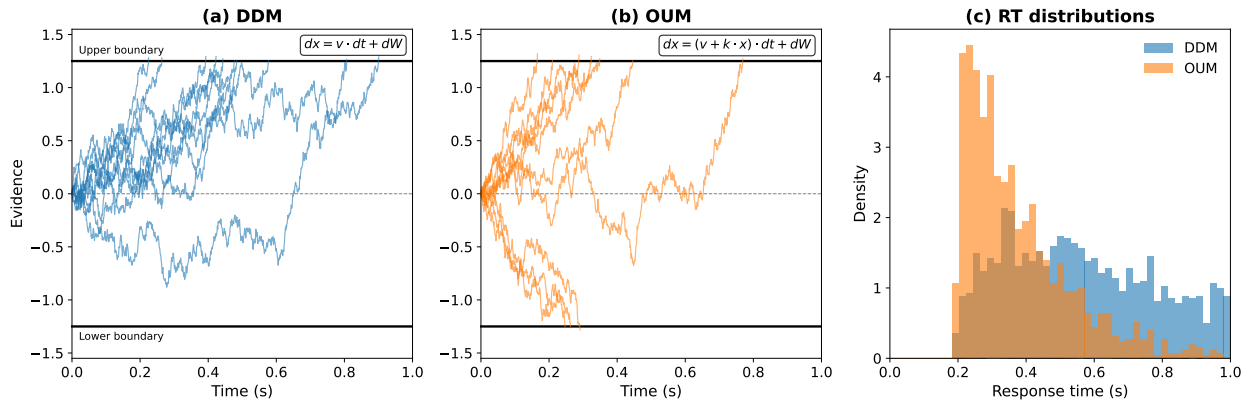

Figure 1

Schematic Illustration of the Diffusion Decision Model and the Ornstein-Uhlenbeck Model

Note. (a) In the standard DDM, evidence accumulates with a constant drift rate ν from a starting point toward one of two decision boundaries separated by a. (b) In the OUM, an additional self-excitation parameter k modulates the drift rate as a function of already accumulated evidence. When $k > 0$, the rate of evidence accumulation accelerates as the process approaches a boundary, creating a feedback loop that speeds up the decision. (c) Resulting RT distributions for the DDM and OUM. Simulation parameters: $\nu = 1.5$, $a = 2.5$, $k = 5.0$, $\tau = 0$.

may not primarily result from lowered decision thresholds, but from an increased tendency to amplify early evidence — a fundamentally different account of a well-established phenomenon.

What might k represent psychologically or neurobiologically? Two broad interpretations seem plausible. First, drawing on the neurophysiological motivation of the original LCA (Usher & McClelland, 2001), self-excitation can be understood as reflecting the ability of neural circuits to amplify and sustain incoming signals. In this view, k maps onto the concept of neural gain — the excitability of neural processing pathways — which is known to be modulated by catecholaminergic (particularly dopaminergic) systems that change across the lifespan (Li & Rieckmann, 2014; Li et al., 2001). Second, at a more psychological level, k might relate to attentional confirmation bias: the tendency to, after

an initial brief screening phase, selectively attend to evidence confirming one's emerging impression (Wang et al., 2024). In the context of S-A tradeoffs, higher k would then reflect a strategic inclination to let early evidence "snowball" rather than sampling all available information exhaustively.

Whether one focuses more on the neuro-physiological or psychological dimension of self-excitation (with both interpretations operating at different levels of description and thus likely not being fully separable) — it seems plausible that a) there are individual differences in k, and b) these individual differences meaningfully relate to other measures that have been employed in cognitive psychology, with age being one of the most prominent. Both the bio-mechanical processes governing natural neural networks and the psychological setups influencing decision making are known to change across the human lifespan. Thus, it seems likely that k is itself not uncorrelated with age. Regardless of whether the OUM ultimately proves to be the superior model for all types of decision tasks, examining whether and how k relates to age should yield new insights into the cognitive dynamics of aging that are invisible within the standard DDM framework. To our knowledge, no previous studies on this topic have been published. We therefore choose the question of whether self-excitation is related to age as our first research question:

RQ1: Is the self-excitation parameter k in the OUM related to age?

In addition, we follow up on the finding that typical parameter patterns found when using a DDM (i.e., boundary separation effects in S-A tradeoffs) did not replicate when an OUM was used instead. Maybe the age effects found for the DDM, particularly in boundary separation, but also in non-decision times, are also not robust when a slightly different model architecture, including self-excitation, is employed? Thus, our second research question focuses on the other parameters of the OUM:

RQ2: How are the drift rate, boundary separation, and non-decision time parameters related to age in the OUM? Is the pattern different from what we find when using a standard DDM?

To approach initial answers to these research questions, we conduct two different studies, each of them focusing on a different aspect essential to understand individual differences in cognition. In both studies, we use state-of-the-art amortized Bayesian workflows, with parameter estimation, model comparison, and validation methodologies based on specialized artificial neural networks as provided in the BayesFlow package (Radev et al., 2023). Utilizing a simulation-based inference approach allowed us to be very flexible in our model formulations, as the only strict precondition we had to meet was to be able to simulate data from our models. As the OUM does not have a closed-form likelihood function, the use of SBI was crucial to conduct our research.

The first study used a wide range of cognitive tasks, as it is known that the age patterns found sometimes differ greatly between experimental setups (Theisen et al., 2021). By analyzing 18 different tasks, we aim to provide first insights on the across-paradigm generalizability of any age patterns in OUM parameters found. In contrast, the second study focuses on one very specific type of task, the Implicit Association Test (IAT; Greenwald et al., 1998). Although this particular task is perhaps not a standard approach in cognitive psychology, the data set used has a unique benefit: the sample size is exceptionally large, with the final N in our study being $5,564,947$. This allows us to analyze in a very fine-grained manner - down to specific years of age - the cross-sectional means of OUM parameters across large parts of the human life span. The use of amortized Bayesian inference with artificial neural networks, where parameter estimation is extremely fast even for very large data sets, enabled us to easily conduct these analyses.

In both studies, we perform our analyses in two steps. First, we use principled model comparisons to provide an answer to the question whether the OUM or the DDM best explain our data. We then proceed by looking at the effects of age on OUM (and DDM) parameters. Is self-excitation related to age in any way? Do the other parameters show any age patterns distinct from what we would expect from what we know about the DDM? Together, our two studies aim to provide a first, but already quite multifaceted,

view on the implications of the self-excitation parameter for studying cognition throughout the human lifespan.

Study 1

In our first study, we focus on the age correlations of OUM model parameters across a wide range of binary decision tasks. As age effects in DDM parameters are known to often differ greatly between different types of tasks (Theisen et al., 2021), it makes sense to start by exploring a broad range of possible experimental paradigms. We use a data set that includes 18 different response time tasks. These data were originally collected in a study focusing on the relationships of drift rates (in the DDM) with intelligence (Lerche et al., 2020). Additionally, there are also published analyses of age effects in DDM parameters based on these data (von Krause et al., 2020), making them an ideal test case for studying age effects in the OUM's parameters. Analysis code and processed data are available at github.com/mischavk/ou_age.

Methods

Sample

The data set includes 125 participants, with 70 of them indicating female, 55 indicating male gender. The mean age of the study was 35 ($SD = 14$), with an age range of 18 to 62. Participants were recruited via a newspaper article, via flyers distributed at public places, and via an online participant pool. Thus, the sample was slightly more diverse than a typical student sample, with only 50% being students. All participants received 35€ as compensation and provided informed consent.

Procedure

Data collection for the entire study was spread across three sessions, of which two included the RT tasks. In each of these two sessions, participants completed nine different RT tasks, with their order being fixed across participants. After the third and sixth task within each session, participants took a three-minute break. In all RT tasks, participants

started with four practice trials to get accustomed to the task. The test block then consisted of 101 trials, with the first trial being discarded as warm-up.

Material

The RT tasks included nine tasks with mean RTs of around 800 ms, which is in the range of response times typically used in evidence accumulation modeling studies. These tasks are labeled *fast tasks* from hereon. In addition, the study included nine more complex RT tasks with mean RTs of around 3 seconds (*slow tasks*). In each case, the nine tasks (*fast* or *slow*) consisted of three tasks each from three different content domains: figural, verbal, or numeric. Table 1 gives an overview of all tasks.

Table 1

Overview of the 3 (domain: figural vs. numeric vs. verbal) × 2 (speed: fast vs. slow) × 3 (number of tasks) = 18 RT tasks.

	Fast	**Slow**
Figural	FF1: dot-rectangle task (1.9)	SF1: maze task (2.1)
	FF2: simple area task (2.4)	SF2: complex area task (1.6)
	FF3: polygon task (1.3)	SF3: pie task (2.7)
Numeric	FN1: number discrimination task (2.2)	SN1: mean value computation task (1.8)
	FN2: odd-even task (1.5)	SN2: equation task (2.5)
	FN3: simple inequation task (2.8)	SN3: complex inequation task (1.2)
Verbal	FV1: word category task (2.6)	SV1: grammar task (1.4)
	FV2: lexical decision task (1.1)	SV2: statement task (2.3)
	FV3: animacy task (1.7)	SV3: semantic category task (2.9)

Note. The first letter indicates the task complexity (F = fast, S = slow); the second letter denotes the domain (N = numeric, V = verbal, F = figural). The numbers in parentheses indicate the time point of assessment (session and number in sequence).

In fast figural tasks, participants needed to detect whether a dot was displayed inside or outside of a rectangle (FF1), which of two rectangles shown was larger (FF2), and whether a polygon shown was a triangle or a rectangle (FF3). In the slow figural tasks, they had to determine if there was a way out of a maze shown (SF1), which out of two three-triangle sets covered the greater total area (SF2), and if three "slices" of a pie plot shown separately added up to more or less than a total circle (SF3). In the fast numeric tasks, participants needed to identify whether a displayed number was smaller or greater than 500 (FN1), whether a number was odd or even (FN2), and which out of two numbers shown was larger than the other (FN3). In the slow numeric tasks, participants saw a cloud of 16 numbers and had to judge whether their average was smaller or greater than 500 (SN2), had to decide whether an equation shown (e.g., $5 * 7 = 35$) was correct or false, and which out of two simple computations led to the greater solution (e.g., 9–6 vs. 19–17). In the fast verbal tasks, participants needed to judge whether a word was an adjective or a noun (FV1), whether the string of letters shown was a word or not a word (FV2), and determine whether the displayed noun related to a living or non-living entity (FV3). Finally, in the slow verbal tasks, participants had to spot a grammatical error in a sentence either in a noun or a pronoun (SV1), had to determine whether a set of words could be combined to form a true sentence (SV2), and needed to judge whether one or two out of a set of five words shown belonged to a different semantic category than the others (SV3). More information on the procedure and detailed descriptions of the tasks can be found in Lerche et al., 2020.

Data preparation and modeling approach

We used all available response time data except for outlier trials that were faster than 200 ms or slower than 10 seconds. As cognitive models, we used, first, a standard DDM, including just the main parameters of drift rate (ν), boundary separation (a), and non-decision time (τ). Second, we used an OUM, that only added self-excitation (k) as an additional parameter. We did not estimate a potential starting point bias but fixed the

starting point to $a/2$, as is standard when using accuracy coding (where the two boundaries represent correct and incorrect responses rather than specific response options), which eliminates the need for a response bias parameter. To keep the model parsimonious and because of the relatively low number of experimental trials per person and task, we fixed all intertrial variabilities of DDM/OUM parameters to zero. This approach has been shown to facilitate the recovery of the main parameters (Lerche & Voss, 2016).

As we used a fully Bayesian approach, we defined broad, but mildly informative priors to use for parameter estimation and model comparison. Our chosen priors had to satisfy several criteria:

1. We needed two sets of priors: one for the fast decision tasks and one for the slow decision tasks, as the empirical data from the two task families were quite different.

2. The priors for both cognitive models and both types of tasks needed to align broadly with standards from the modeling literature (Matzke & Wagenmakers, 2009). For example, drift rates > 10 should be extremely unlikely.

3. The prior predictive distributions of correct and error RTs and accuracy rates needed to roughly correspond to the empirical data in the tasks. I.e., they needed to imply mean RTs of about $500 - 1000$ms in the fast tasks ($90 - 95\%$ accuracy), and about 2-3 seconds ($80 - 95\%$ accuracy) in the slow tasks.

4. The prior predictive distributions needed to be similar for the two cognitive models, to allow for a fair prior predictive model comparison (that is not won by one model simply because it implies faster mean RTs than the other).

5. The priors needed to be chosen such that the confusion matrix of the model comparison was unbiased. We aimed for a maximum $< .1$ difference between confusion rates (i.e, the classifier being better at correctly identifying one of the cognitive models than the other).

6. The priors needed to be chosen such that recovery (correlations between true and estimated parameters in simulated data) was adequate for all parameters.

Table 2 shows our chosen priors for both the DDM and the OUM based on these criteria. We chose gamma priors for drift rates to cover the typical ranges found in

Table 2

Prior distributions for DDM and OUM parameters.

Parameter	Study 1 / Study 2	DDM	OUM
Fast tasks (Study 1) and IAT (Study 2)			
ν, ν_1, ν_2		$\text{Gamma}(3.5, 1.0)$	$\text{Gamma}(3.5, 1.0)$
a, a_1, a_2		$8 \cdot \sigma(\mathcal{N}(-1.0, 0.7))$	$8 \cdot \sigma(\mathcal{N}(0.3, 1.0))$
τ, τ_c		$\text{U}(0.1, 1.0)$	$\text{U}(0.1, 1.0)$
τ_e	Study 2 only	$\text{Gamma}(2.0, 0.3)$	$\text{Gamma}(2.0, 0.3)$
k		—	$\text{Gamma}(4.0, 0.5) + 1$
Slow tasks (Study 1)			
ν		$\text{Gamma}(6.0, 0.25)$	$\text{Gamma}(6.0, 0.25)$
a		$10 \cdot \sigma(\mathcal{N}(-1.3, 1.0))$	$10 \cdot \sigma(\mathcal{N}(1.3, 2.0))$
τ		$\text{U}(0.1, 3.0)$	$\text{U}(0.1, 3.0)$
k		—	$\text{Gamma}(4.0, 0.5) + 1$

Note. $\sigma(\cdot)$ denotes the logistic sigmoid function, so $a = L \cdot \sigma(\mathcal{N}(\mu, \sigma^2))$ yields a smoothly bounded prior on $(0, L)$. Gamma distributions are parameterized as $\text{Gamma}(\text{shape}, \text{scale})$. In Study 1, each task has a single drift rate (ν), boundary separation (a), and non-decision time (τ). In Study 2, subscripts 1/2 denote the congruent/incongruent conditions; τ_c and τ_e denote non-decision times for correct and error responses, respectively. The error non-decision time τ_e is specific to Study 2.

previous experimental studies, with a higher mean for fast tasks (where drift rates tend to be higher) than for slow tasks. For boundary separation, we used sigmoid-bounded priors of the form $a = L \cdot \sigma(\mathcal{N}(\mu, \sigma^2))$, where $\sigma(\cdot)$ is the logistic sigmoid function. This yields a

smoothly bounded prior on $(0, L)$ that avoids hard uniform boundaries while still constraining the parameter to a plausible range. We used different sigmoid parameters for the DDM and OUM to account for the fact that OUM boundary separation estimates tend to be higher than DDM estimates (Wang et al., 2024), and different upper bounds L for fast ($L = 8$) and slow ($L = 10$) tasks. For non-decision times, we used uniform priors with wider ranges for slow tasks to accommodate their longer non-decisional components. For self-excitation, we chose a shifted gamma prior, $k \sim \text{Gamma}(4.0, 0.5) + 1.0$, which restricts k to positive values above 1.0. This meant that the parameter did not track potential leakage effects (i.e., a pull back to the starting point with increasing evidence strength), and was also distinct from the DDM (where k is 0), which was importing for conducting the model comparison.

Model Comparison

We used a simulation-based Bayesian prior predictive model comparison procedure as implemented in the `BayesFlow` 2.0.10 Python library (Radev et al., 2021). In this approach, a classification network is trained on simulations from multiple (in our case, two) generative models. The trained network can then be used to approximate the true posterior model probabilities for new (empirical) data. This is possible regardless of whether a closed-form likelihood function is available for all models, which is crucial when comparing DDM and OUM.

We used $32,000$ simulations from both models and trained our neural networks (a combination of a set transformer summary network compressing the input data into 20 summary dimensions and the classification network) for 100 epochs. We assumed equal prior model probabilities. Visual inspection of the calibration patterns and the confusion matrix (see Figures A1 and A2 in the Appendix) indicated adequate calibration and good discernibility of the two cognitive models in $10,000$ (separately simulated) validation data sets.

Parameter estimation

To approximate the joint posterior distributions of cognitive parameters in the DDM and OUM, we used the simulation-based inference method implemented in `BayesFlow` 2.0.10 (Radev et al., 2020, 2023). We trained separate networks for each of the two models. In each case, we generated $64,000$ synthetic training data sets consisting of evidence accumulation model parameters and 100 trials of response time data generated from each of these individual parameter sets as well as 1000 validation data sets. The neural networks then learned the associations between the simulated RT data and the respective parameters of the cognitive models. We used a set transformer summary network compressing each participant's RT data (for a given task) into a 20-dimensional summary, and a coupling flow as our inference network; we trained the networks for 100 epochs.

As our recovery studies showed, the networks were able to adequately recover the true data-generating parameters in 500 newly simulated data sets (see Figures A3–A6). However, recovery for the boundary separation parameter was worse in the OUM compared to the DDM. We discuss this finding later and try to account for it in our statistical analyses. Analyses of computational faithfulness/simulation-based calibration revealed no global biases in either posterior location or dispersion, suggesting that posterior credibilities were well-calibrated for all cognitive parameters estimated.

After training the neural networks, we obtained $3,000$ samples from the joint approximate posterior distribution separately for each cognitive model, person, and RT task. We discarded any posterior samples with theoretically impossible values, that is, negative boundary separations and negative non-decision times. The relative number of such posterior samples was low (DDM: 0.17%; OUM: 0.40%, across all 18 tasks and both parameters). We did not discard negative samples for drift rates and self-excitation, as these values are theoretically possible (although beyond the range of our chosen priors) and to be expected, giving the typical Gaussian distributions of parameter posteriors and posterior means sometimes quite close to zero.

Finally, we conducted posterior predictive checks to compare model fit of DDM vs. OUM in this metric. These analyses are reported in the Appendix (B1, B2).

Statistical Analyses

After obtaining estimates of DDM and OUM parameters, we computed, from the full joint posterior distributions, simple Pearson age correlations for each cognitive parameter.

Results

Model comparison

Figure 2 shows the distributions of approximate posterior model probabilities for the DDM and OUM across all 18 RT tasks. A consistent pattern emerges that differentiates by task speed. The OUM tends to be favored over the DDM in the majority of fast RT tasks (mean RTs $<$ 1 second). Conversely, the DDM is strongly preferred for the slower, more complex tasks (mean RTs $>$ 2 seconds). There are no obvious differences between different content domains (figural, numeric, verbal). We note that prior predictive model comparison outcomes can be sensitive to the specific prior distributions chosen and therefore interpret these results as suggestive rather than definitive, especially given the sometimes small differences in model probabilities for the fast tasks. The overall pattern — a dissociation by task speed — is, however, quite consistent.

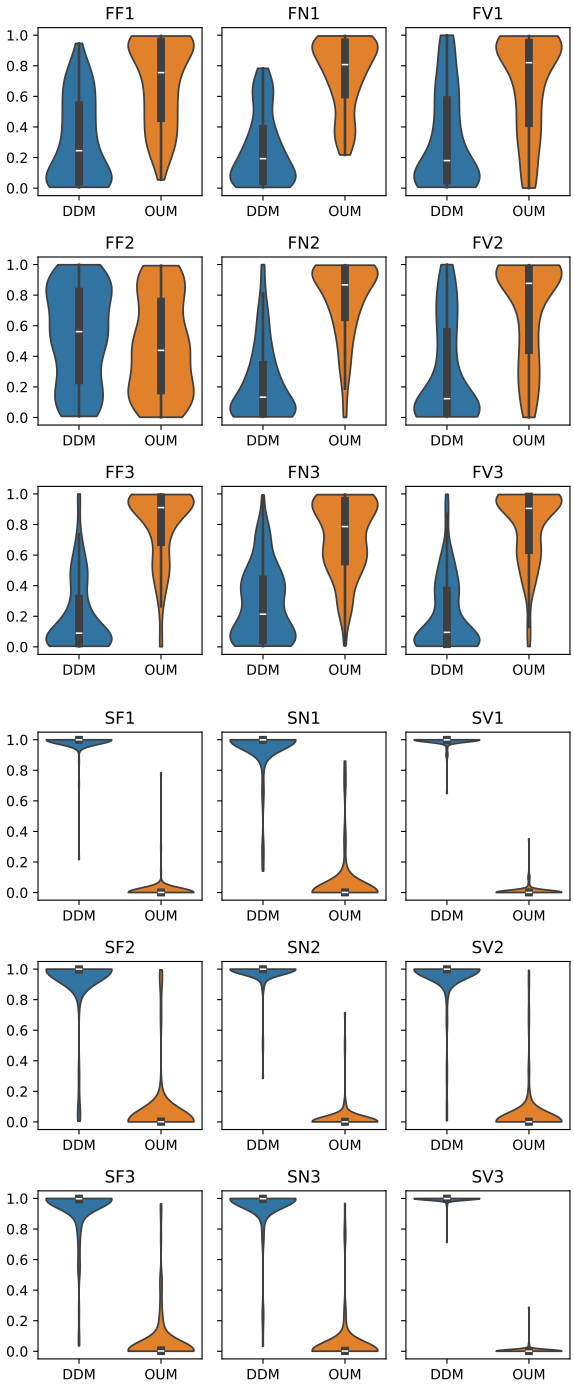

Figure 2

Posterior Model Probabilities for the DDM and OUM Across 18 RT Tasks

Note. Distributions of approximate posterior model probabilities for the DDM (blue) and OUM (orange) for each of the 18 RT tasks. Top panel: Fast tasks (mean RT < 1 s). Bottom panel: Slow tasks (mean RT > 2 s). Each subplot shows one task with violin plots for the two models. The OUM tends to be favored in the majority of fast tasks, while the DDM is strongly preferred for the slower, more complex tasks.

Age correlations

Figures 3 and 4 show our results on age correlations in the 18 RT tasks. For every task, we plot the posterior correlations of participant ages with drift rates (ν), boundary separations (a), non-decision times (τ), and, in case of the OUM, self-excitation (k). Different colors indicate the two different cognitive models. Inspection of the age correlations of the k parameter shows a robust pattern across all 18 RT tasks. k exhibits a small to moderate negative age correlation; older participants tend to have lower self-excitation estimates. This pattern also holds for the slow RT tasks (lower half of the figure), for which the OUM showed worse fit than the DDM in our model comparison.

The general patterns of age correlations of the other parameters in the OUM largely mirror those found for the DDM. This holds across both fast and slow tasks and across all three content domains. Especially for the drift rate parameter, posterior distributions of age distributions are very similar across the two models, with large parts of the posteriors close to zero in the majority of tasks. However, for boundary separations and non-decision times, a slightly more nuanced pattern emerges. First, the posterior distributions exhibit more spread for the OUM estimates, indicating more uncertainty, probably due to possible trade-offs between k and a. Second, for both the OUM and the DDM, age is positively correlated to boundary separations and non-decision times across the majority of tasks. However, in the OUM, age correlations in both parameters tend to be lower in both some fast and some slow tasks, although in many cases the posterior distributions still show much overlap across models. This attenuation suggests that part of what the DDM attributes to increased decision caution and slower non-decisional processing in older adults may instead reflect age-related changes in self-excitation.

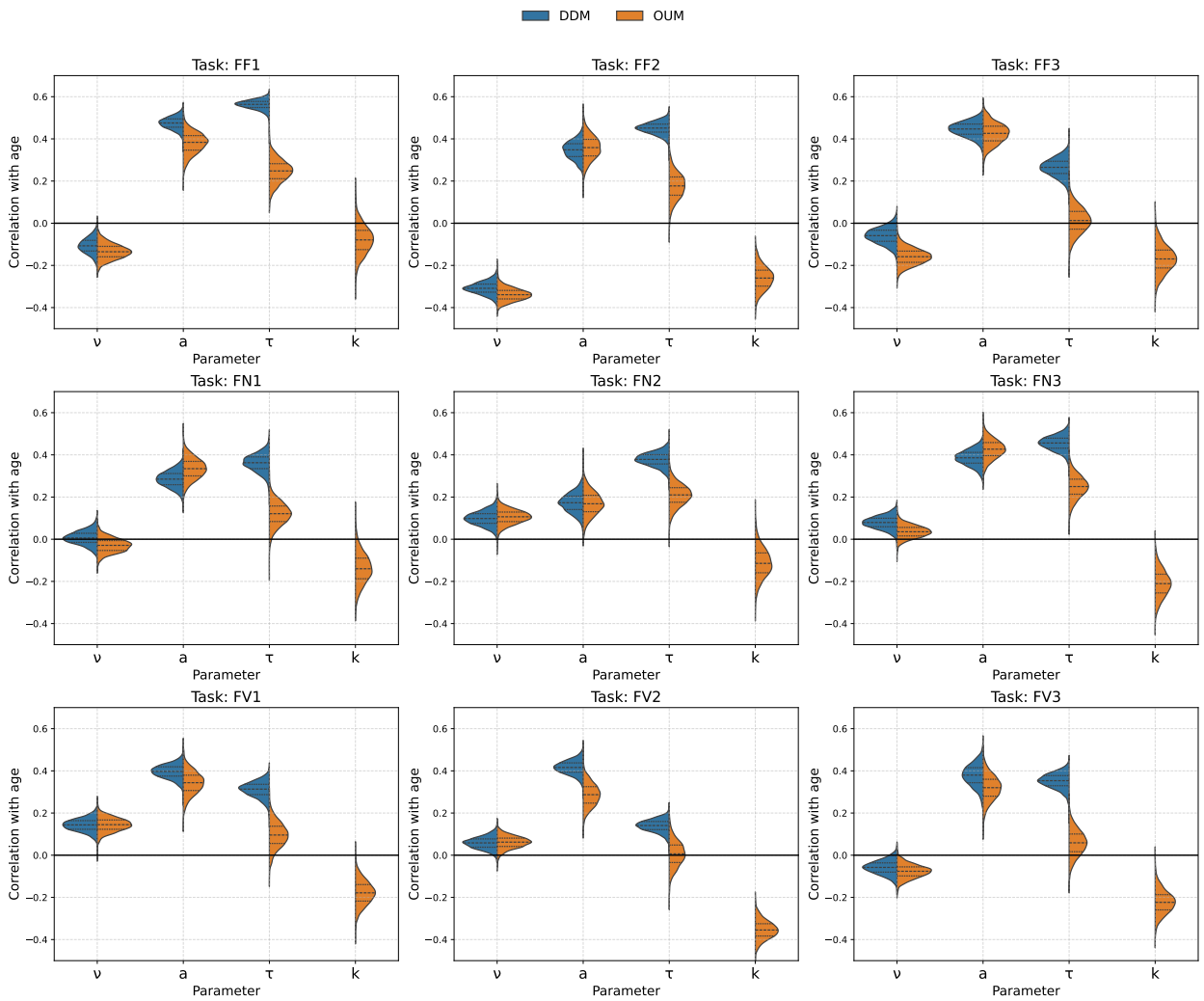

Figure 3

Posterior Distributions of Age Correlations With DDM and OUM Parameters Across 9 Fast RT Tasks

Note. Posterior distributions of Pearson correlations between participant age and cognitive model parameters for fast tasks. Different colors indicate the DDM and OUM. For each task, correlations are shown for drift rate (ν), boundary separation (a), non-decision time (τ), and, for the OUM, self-excitation (k). The self-excitation parameter shows a small negative age correlation across all tasks, while boundary separation and particularly non-decision time age correlations tend to be slightly attenuated in the OUM compared to the DDM.

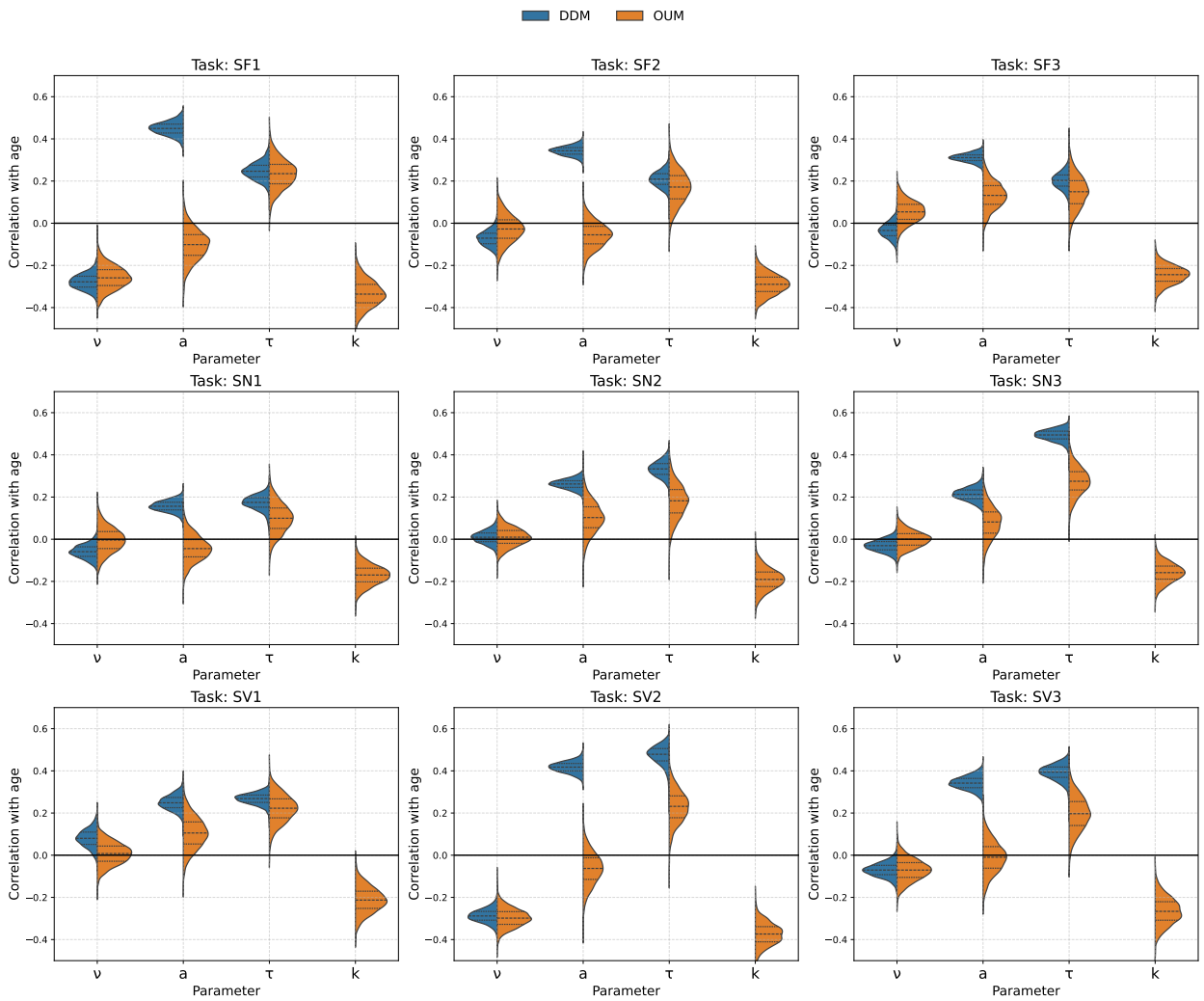

Figure 4

Posterior Distributions of Age Correlations With DDM and OUM Parameters Across 9 Slow RT Tasks

Note. Posterior distributions of Pearson correlations between participant age and cognitive model parameters for slow tasks. Different colors indicate the DDM and OUM. For each task, correlations are shown for drift rate (ν), boundary separation (a), non-decision time (τ), and, for the OUM, self-excitation (k). The self-excitation parameter shows a small negative age correlation across all tasks, while non-decision time and particularly boundary separation age correlations tend to be attenuated in the OUM compared to the DDM.

Study 2

In our second study, we focus on analyzing detailed cross-sectional age patterns in OUM parameters, especially the self-excitation parameter k, in a very large data set. We use data from over five million participants who participated in an online implicit association test (IAT) experiment (Xu et al., 2014). In this way, we move from studying age effects across different tasks to studying parameter patterns in very specific age ranges, as well as across different demographic subgroups.

Methods

Sample

We used publicly available Race IAT data from Project Implicit (Xu et al., 2014)). We obtained all raw response time data available through the project's OSF site (https://osf.io/52qxl) for the years 2003 to 2023. In this time frame, $9,612,305$ people completed the race IAT via the Project Implicit system. Out of these, $6,127,948$ participants (100% related to the following numbers) also indicated their age, our main covariate of interest. As data collected online with limited experimental control can be expected to contain noise, we cleaned the data in several steps. First, we excluded participants with invalid session IDs ($> 99.99\%$ remaining), response latencies saved as zero (99.6% remaining), and missing trials (97.9% remaining). Next, we excluded participants with below chance accuracy (97.7% remaining) and no error trials (91.5% remaining). The latter step was necessary in order to identify all DDM and OUM parameter, as described below. Further, we excluded careless responders; we chose a within-person criterion of more than 20% response times faster than 200 ms or slower than 10 seconds (91% remaining). As there were very limited data from people indicating an age of less than 10 years or more than 80 years, we used this age range for our analyses, leading to a final sample size of $5,564,947$. 60% of our participants reported a female birth sex. The mean age was 29.39 ($SD = 13.05$). 68% reported at least some college education, indicating that our sample was over-educated compared to the general population.

Procedure

The exact experimental procedure is described on the Project Implicit OSF site. All participants completed a standard IAT, consisting of five different experimental blocks. For our analyses, we used (as is standard in the literature) the trials from the so-called congruent and incongruent test blocks, each containing 60 trials. The amounts of trials in the practice blocks (which we did not use) varied over the years of data collection as described on the Project Implicit OSF site.

Materials

The race IAT (Greenwald et al., 1998, 2003, 2015) is by now a quasi-standard task in social psychology, although its interpretation is still controversial (Fiedler et al., 2006; Schimmack, 2021) and its use is less common in cognitive psychology. In each experimental trial, participants classify either a word or an image stimulus as belonging to one of two categories. In the test blocks, both picture stimuli and word stimuli are shown in turn. The word stimuli (e.g., "love" or "horrible") belong either to the category "good" or to the category "bad". The picture stimuli show human faces, and have to categorized either as African American or European American. [1] In the so-called *congruent* experimental condition, the categories "good" and "European American" (vs. "bad" and "African American") map onto the same response button, while the reverse pattern applies for the *incongruent* condition.

While the task is typically used to study implicit racial biases by computing the weighted difference in response times between the two experimental conditions, we did not focus on this interpretation. Instead, we used the IAT simply as an instantiation of a binary decision task, that has also in the past been analyzed successfully with the DDM (Klauer et al., 2007; von Krause et al., 2021, 2022).

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[1] Over the course of data collection (2003–2023), the exact racial category labels used in the Race IAT were updated several times. The core task structure remained unchanged. Details are available on the Project Implicit OSF site.

Data preparation and modeling approach

We excluded all RT trials faster than 200 ms or slower than 10 seconds (as we had already done in the first study). As our cognitive models, we once more used a simple DDM containing just the main parameters of drift rate, boundary separation, and non-decision times, and an OUM that added the self-excitation parameter k. In both models, we estimated separate drift rates and boundary separations for each of the two experimental conditions. Also in both models, we estimated two separate non-decision times for correct responses and error responses. This was necessary to account for the specific way error responses are recorded in Project Implicit: the saved error response time includes not only the time needed to give an initial (erroneous) response, but also the time needed to correct this response and hit the keyboard button mapping the correct answer. In our modeling framework, the error non-decision times also contain this additional "correction" time, which we assume to be independent from the original decision process. As in Study 1, we chose a fully Bayesian procedure to conduct our model comparison and parameter estimation. We used the same prior distributions as for the fast tasks in Study 1 for the corresponding parameters (see Table 2). We used a Gamma$(2.0, 0.3)$ prior for the error non-decision time to account for the additional correction time included in IAT error RTs (similar to previous studies with the same data set; von Krause et al., 2022).

Model comparison

Once more, we used the prior-predictive approach to model comparison as implemented in `BayesFlow`. We used $32,000$ simulations from both cognitive models (DDM and OUM) to train a classifier network to approximate the posterior model probabilities (assuming equal prior probabilities). Once more we used a set transformer summary network to compress the input data into 20 summary dimensions and trained the networks for 100 epochs. Calibration plots and confusion matrix (shown in Figures D1 and D2 in the Appendix) indicated adequate convergence. In addition, the calibration plot did not indicate any strong bias in favor of either the DDM or OUM.

Parameter Estimation

To obtain parameter estimates, we once more chose amortized Bayesian inference. Given the extraordinary sample size, this would have been the only feasible option even if we had not used a cognitive model without an analytical likelihood function like the OUM. We trained separate approximators for the two cognitive models. In each case, we used $64,000$ simulations (plus 1000 for validation) and trained the networks for 100 epochs. We used set transformer summary networks (with 20 summary dimensions) and coupling flow inference networks.

We simulated an additional 500 data sets to use for simulation-based calibration and recovery checks. The ECDF plots indicated satisfactory calibration of the approximate posteriors across all parameters in the two cognitive models (see Figures D3 and D4 in the Appendix). However, similar to Study 1, the *in silico* correlation of true parameters and parameter estimates as obtained from the networks was slightly lower for the boundary separation parameters in the OUM, compared to the DDM. For all other parameters, recovery was adequate - some error non-decision times could not be recovered as the respective data sets contained no or only extremely few error trials.

After training the neural networks, we obtained 3000 posterior samples for each empirical data set for each of the cognitive models. Also similar to Study 1, we discarded all samples with theoretically impossible parameter values (i.e., negative boundary separations or non-decision times). Finally, we conducted posterior predictive checks to compare model fit of DDM vs. OUM in this metric. These analyses are reported in the Appendix E1.

Statistical analyses

Given the very large sample size and exploratory nature of the study, we chose simple, descriptive analyses. For each parameter of the cognitive models, we computed the across-person means and across-person standards deviations for each year of age between 10 years and 80 years. We then plotted the results. We repeated this procedure across different demographic subgroup, splitting by gender, education, and geographic location.

Results

Model Comparison

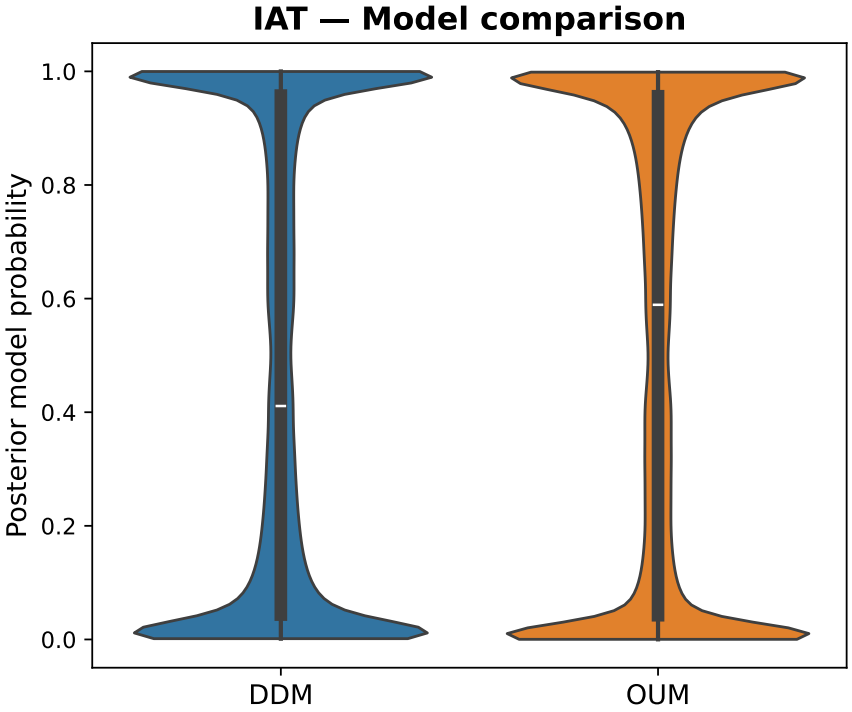

Figure 5

Posterior Model Probabilities for the DDM and OUM in the IAT Data (Study 2)

Note. Violin plots of approximate posterior model probabilities across all participants in the Race IAT. The overall trend favors the OUM, with a median posterior model probability of .59.

Figure 5 shows the results of our model comparison for the IAT data. We plot the across-person distributions of posterior model probabilities for the DDM and OUM. There are substantial individual differences in model fit: for a significant minority of the participants $(2,609,980)$, the DDM fits better, while for the rest $(2,966,158)$, the OUM is the better model. However, the overall trend favors the OUM, with a median posterior model probability of .59 across participants (vs. .41 for the DDM). As with Study 1, these results are to some extent dependent on our chosen prior distributions. Nevertheless, we take the overall pattern — combined with the consistent results for fast tasks in Study 1 —

as sufficient motivation to examine the OUM's parameters, and k in particular, in these data.

Age trends

Figures 6 and 7 show the results of our analyses of age trends in the DDM and OUM parameters.

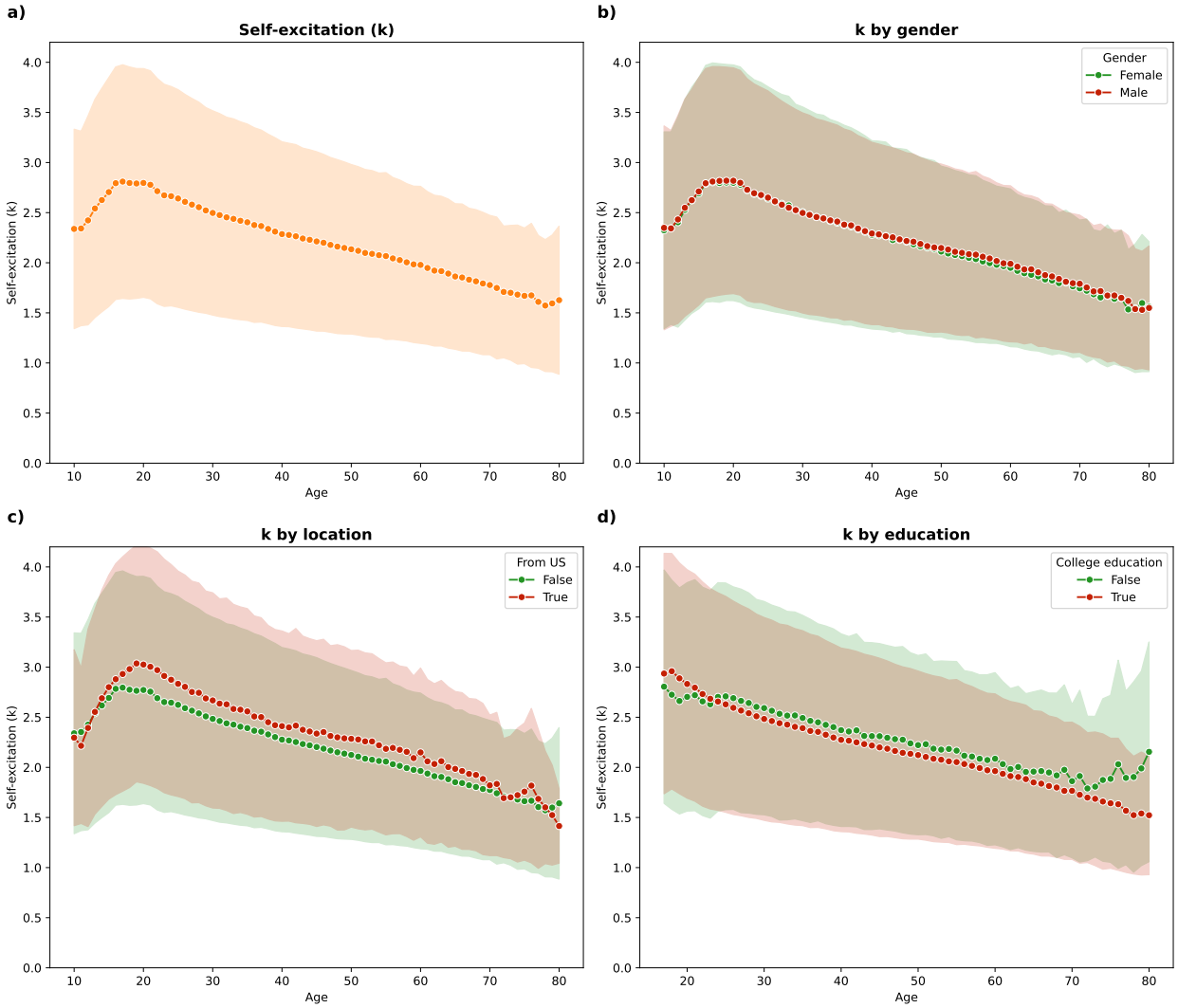

Figure 6

Cross-Sectional Age Trends in Self-Excitation (k) Across Demographic Subgroups (Study 2)

Note. Across-person means (± 1 SD) of the OUM self-excitation parameter k for each year of age from 10 to 80. (a) Overall sample. (b) Split by gender. (c) Split by geographic location (US vs. non-US). (d) Split by education level (college education vs. no college education; restricted to ages ≥ 17).

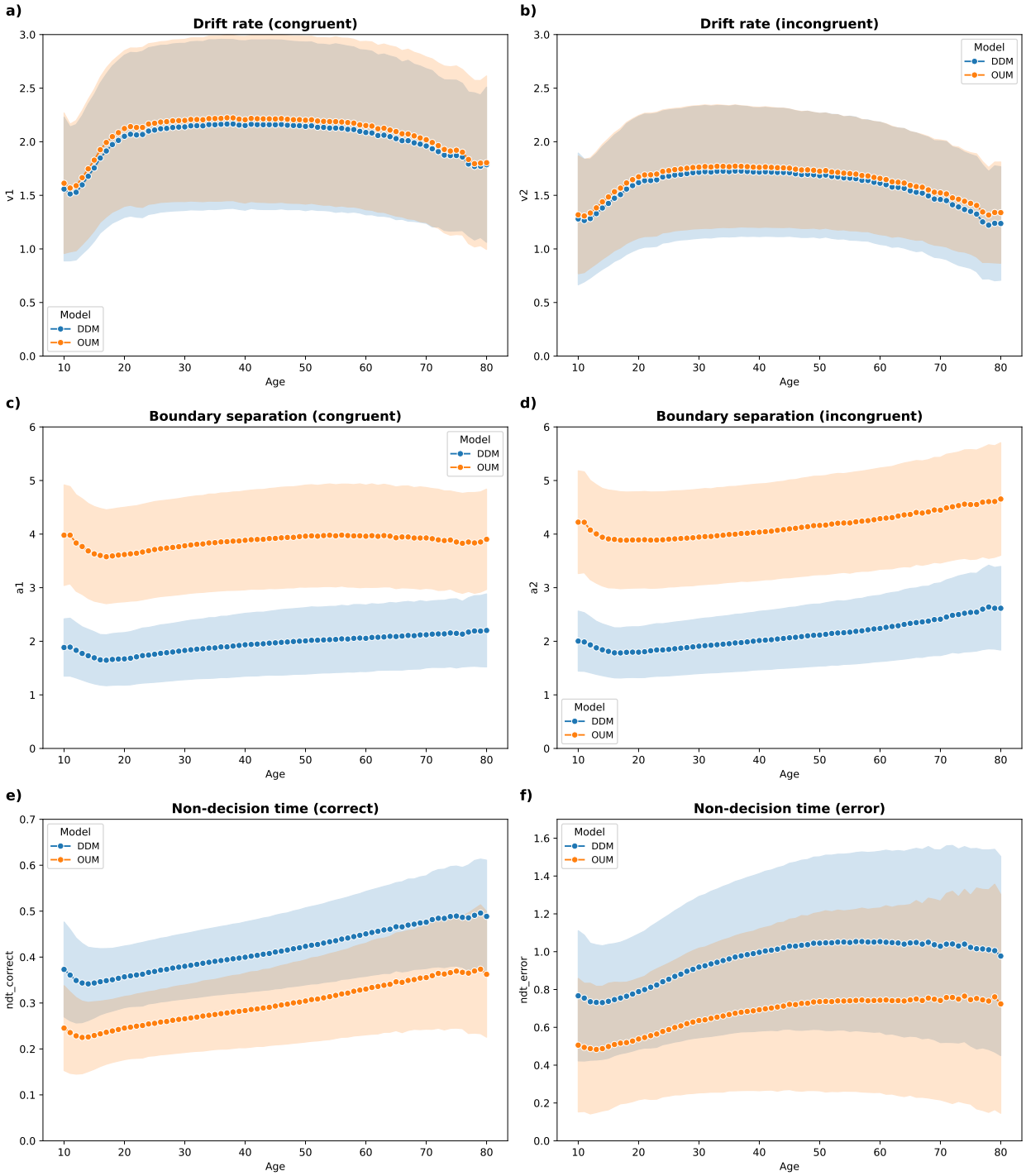

Figure 7

Cross-Sectional Age Trends in DDM and OUM Parameters in the IAT Data (Study 2)

Note. Across-person means (± 1 *SD*) of DDM (blue) and OUM (orange) parameter estimates for each year of age from 10 to 80. (a) Drift rate, congruent condition. (b) Drift rate, incongruent condition. (c) Boundary separation, congruent condition. (d) Boundary separation, incongruent condition. (e) Non-decision time, correct responses. (f) Non-decision time, error responses.

Figure 6a shows, for each year of age between 10 and 80, the across-person means and standard deviations in the self-excitation parameter k that is specific to the OUM. Cross-sectional mean values increase from age 10 to age 17. After a plateau that lasts until age 21, self-excitation then shows a consistent quasi-linear (cross-sectional) decrease throughout the rest of the studied age period. For all years of age, mean values are in the range of 1.57 (for age 78) to 2.81 (for age 17), with a mean standard deviation of 0.90. Cohen's d for the difference between these two extreme categories is 1.31. Sub-group analyses showed very little differences regarding the age dynamics of k, although there were small mean level differences between groups.

Drift rates in both experimental conditions showed an inverted-U shaped trajectory across the age range studied. Mean drift rates increased until approximately age 25–30, remained roughly stable through middle adulthood, and began to decline after age 55–60 — with the decline starting somewhat earlier in the incongruent condition (around age 50). The age trajectories were highly similar across the two cognitive models, both in shape and in absolute values. In the incongruent condition, OUM drift rate estimates were slightly higher on average than DDM estimates (Cohen's $d = 0.09$), while congruent drift rates were virtually identical across models. Across person variability in drift rates was comparable between the DDM and OUM (SDs of 0.77 vs. 0.75 in the congruent condition; SDs of 0.61 vs. 0.57 in the incongruent condition).

Boundary separation estimates in both experimental conditions showed a U-shaped pattern, decreasing from age 10 to approximately age 18, then increasing across the remainder of the studied age range. This pattern was qualitatively identical across cognitive models and experimental conditions. However, the OUM yielded substantially higher mean boundary separations than the DDM (Cohen's $d = 2.58$ in the congruent condition; 2.76 in the incongruent condition), accompanied by greater across-person variability (SDs of 0.91 in the OUM vs. 0.53–0.55 in the DDM). The inflation of boundary separation estimates in the OUM likely reflects a partial trade-off with the self-excitation

parameter k, a point we return to in the Discussion.

Non-decision time estimates for both correct and error responses showed a U-shaped trajectory, decreasing until approximately age 13–14, then increasing with age. In both cases, the age-related increase was more pronounced in the DDM than in the OUM. This difference was particularly striking for correct non-decision times, where the DDM yielded substantially higher mean estimates (Cohen's $d = 1.42$) and a stronger age correlation ($r = .38$ in the DDM vs. $r = .34$ in the OUM), while across-person standard deviations in correct non-decision times were 0.08 in both models. For error non-decision times, the model differences were smaller (Cohen's $d = 0.71$) and variability was virtually identical across models ($SD = 0.39$). As with boundary separation, the attenuated age effects in the OUM suggest that the standard DDM may partly misattribute age-related variance in k to the non-decision time parameters.

Table 3

Pearson age correlations for DDM and OUM parameters in the IAT (Study 2, $N = 5564947$).

Parameter	DDM	OUM
Drift rate (congruent, v_1)	+0.068	+0.062
Drift rate (incongruent, v_2)	+0.051	+0.048
Boundary separation (congruent, a_1)	+0.237	+0.135
Boundary separation (incongruent, a_2)	+0.261	+0.129
Non-decision time (correct, τ_c)	+0.377	+0.339
Non-decision time (error, τ_e)	+0.263	+0.199
Self-excitation (k)	—	−0.242

Note. Correlations are computed from the median of the posterior parameter estimates per participant, restricted to ages 10–80.

Table 3 summarizes the Pearson age correlations across all parameters for both

models. In all cases, age effects that were present in one model were also present in the other, with the exception of k, which is unique to the OUM. However, the magnitudes of age correlations for boundary separation and non-decision time were slightly smaller in the OUM than in the DDM.

Discussion

In this study, we provide a large-scale investigation into the Ornstein-Uhlenbeck model (OUM) of decision making as a mechanistic framework for understanding cognitive aging. To this end, we integrate data from a battery of 18 cognitive tasks (Study 1) with a massive 5.5 million participant implicit association test (IAT) data set (Study 2). Our core empirical discovery is the identification of a systematic, quasi-linear decline in the OUM's self-excitation parameter (k) across the adult lifespan. k reached its maximum in late adolescence, followed by a steady decrease starting in early adulthood. Model comparisons tended to favor the OUM over the DDM for most fast binary decision tasks (with mean RTs $<$ 1s), including the IAT, while the DDM remained the preferred model for more complex tasks with slower mean RTs ($>$ 2s). Interestingly, in the IAT, there was great across-person heterogeneity regarding which of the two models fit best. In the OUM, accounting for self-excitation attenuated the magnitude of age effects on the boundary separation and non-decision time parameters. Our results suggest that age-related slowing of response times in simple binary decision making might not merely be a product of increased decision caution and a slowdown of non-decisional processes, but also of an age-related reduction in the self-sustaining momentum of evidence accumulation — a dynamic overlooked in previous modeling approaches.

In the OUM, the model's k parameter creates a feedback loop where the current state of accumulated evidence influences the current rate of evidence accumulation — in contrast to the standard DDM, which assumes constant drift rates within an experimental trial. If $k > 0$, a self-excitatory process emerges: initial evidence creates a "pull" towards one decision boundary that grows ever stronger as the boundary is approached (Busemeyer

& Townsend, 1993). This state-dependency of the accumulation process might be understood as the system's internal drive to conclude the decision, and is principally unrelated to the drift rate (mapping, in individual differences studies, the overall efficiency of evidence accumulation) and boundary separation (mapping decision caution). Our results suggest a transition from strongly self-excitatory dynamics in young adulthood towards more constant drift rates in old adults: with increasing age, the decision system might become less sensitive to its own internal state and more reliant on external stimulus strength alone. In the following, we discuss some possible interpretations of these age patterns in k.

One possible interpretation is grounded in the framework suggested by Wang et al. (2024), who suggested that k might be a proxy for attentional confirmation bias. In their study of speed-accuracy tradeoffs, they found that speeded decisions were not achieved primarily by lowering the decision threshold, but by increased self-excitation to amplify early evidence. This could be due to a mechanism in which, once an initial lead for an option is established, *attention* is selectively biased toward features that confirm that choice. For instance, in a color discrimination task, with a stimulus appearing slightly "blue" initially, high self-excitation might reflect an observer's tendency to focus exclusively on blue features, actively ignoring contradictory orange features to speed up the choice. The fact that in our study older participants showed, in the majority of tasks, lower self-excitation, might thus be interpreted as them either lacking the capacity or the strategic inclination to engage in this rapid, bias-driven attentional filtering. This would suggest that older adults, with on average lower ks, may process information more objectively by resisting confirmation bias and sampling the stimulus evenly throughout the entire duration of the trial. If this interpretation is correct, older adults might be less susceptible to early noisy or misleading evidence. However, the penalty for this objectivity would be a loss of speed. If seen in this way, k could even be understood as a second measure of decision caution, independent of boundary separation: older people are slower

because they avoid attentional filtering, thus prioritizing exhaustive evidence over heuristic speed.

Whether or not these differences in attentional filtering found between young and old adults reflect strategic adjustments (as can be expected for the speed-accuracy-tradeoff studies) remains an open question. The original LCA, on which the OUM of decision making is based, was explicitly conceptualized as a model incorporating knowledge about neurophysiological processes, with self-excitation being one of the most important properties of natural neural networks. A neural network's ability to amplify and sustain incoming relevant information is crucial for fast processing. Given the neural and neuro-modulatory developments occurring over the course of the human lifespan, one can assume an influence of these age-related changes occurring at the basic neurophysiological level.

Older adults are known to have on average lower dopamine levels than younger adults, and this difference in neuromodulatory systems has been suggested as a possible factor explaining cognitive aging (Li & Rieckmann, 2014; Li et al., 2001, 2021). Catecholamines (dopamine and noradrenaline) modulate the "gain" or excitability of neural circuits and are thus thought to modulate the "signal-to-noise-ratio" in neural circuits (Servan-Schreiber et al., 1990) . Lower "gain" could in turn lead to "blurry" neural representations, hindering quick decision making (Koen & Rugg, 2019; Koen et al., 2020). Similarly, age-related loss of D1/D2 receptors could weaken the recurrent feedback loops that support neural self-excitation (Bäckman et al., 2006). Following this line of research, self-excitation or "gain" might thus be understood primarily as beneficial for the adequate and timely integration of information. Lower self-excitation would then represent a lower ability to focus filter the relevant information, a failure of the active selection mechanism required to suppress noise. Reduced dopamine might explain why these changes in dynamical neural gain — and thus k — occur in older adults. However, given the purely behavioral data used in the present study, we can, of course, at this point not give any

empirical tests of these interpretations. Future pharmacological challenge studies could, for example, administer L-DOPA to older adults to see if it temporarily restores k (self-excitation), thereby causally linking dopamine to (neural) self-excitation.

Regarding the other cognitive parameters, interestingly the age correlations and age differences (Cohen's ds) of non-decision times and/or boundary separations were often slightly lower in the OUM compared to the DDM. This was true for majority of the fast RT tasks, including the IAT. Given that the OUM fit these data sets slightly better than the DDM, previous studies utilizing the DDM for comparable tasks might have overestimated age-related increases in boundary separation and non-decision times by failing to account for the decline in self-excitation, as high k can empirically largely resemble a low boundary separation (a). The fact that the age effects in non-decision time were often smaller in the OUM, although recovery was similarly good for this parameter in both models, model suggests that the differences in boundary separation should also not just be an artifact of lower reliability in the OUM. Importantly, in our studies the age differences in boundary separation and non-decision times did not vanish for most tasks, although their absolute values tended to be lower and posteriors less precise. In this way, our analyses suggest that older adults tend to be slower because they are both more cautious *and* because their evidence accumulation lacks the internal momentum typical for younger participants. k and a relate to two different aspects of decision making: while boundary separation is conceptualized as a strategic state which is determined pre-decisionally, self-excitation is a dynamic, intra-decisional process.

A strength of our study is the inclusion of a wide range of cognitive tasks, particularly the fact that we also analyzed a set of more complex RT tasks with mean RTs < 2s. While the OUM tended to be the better model for explaining the empirical data in the fast tasks resembling the typical types of tasks used in previous evidence accumulation modeling studies, the DDM remained the better model in the slow, more deliberative tasks. Self-excitation thus seems to be particularly important for understanding rapid, largely

automatic processing. For the slow tasks, k values were generally low. In these more complex tasks, k may be inhibited to allow for exhaustive integration and avoid premature error based on incomplete evidence. In contrast, the OUM was the better model for the majority of participants in the IAT, implying that self-excitation may be an important mechanism for understanding such data. However, despite differences in model fit, the age-k correlation was stable across paradigms, indicating a general trend.

Our study has several important methodological strengths. At the heart of our methodological approach lies the use of simulation-based inference through the `BayesFlow` library, allowing us to apply a fully Bayesian pipeline of model validation, model comparison, and model application (inference). Our SBI approach enabled us to use and vigorously validate the OUM, a model without a known closed-form likelihood function. It also made it possible to easily apply the OUM (and a standard DDM) to the 5.5 million participant IAT data set with relatively low computational cost. This allowed for a precise mapping of the developmental peak and subsequent decline of k across the adult lifespan, at least for this particular paradigm. While parameter recovery between k and boundary separation remains a challenge, the convergence of results across 18 tasks and millions of participants provides high confidence. Our approach is part of a general trend towards SBI-based *Big Data* approaches in the cognitive sciences to validate complex, theoretically rich psychological models by embracing the tools of *behavioral data science*.

Naturally, our study also comes with several limitations and constraints. Both data sets we use are cross-sectional, thus not allowing any conclusions about actual longitudinal developmental processes. In our analyses we are, accordingly, not able to definitely separate intra-individual aging from cohort effects. Generational differences in “speed-focus” (e.g., digital native status) could potentially influence estimates of k independently of biological aging. Longitudinal studies of cognitive aging often show smaller effect sizes compared to cross-sectional data, although the general trends typically stay intact (Zimprich & Martin, 2002). It must be noted that the Study 2 sample, while exceptionally large, was drawn

from a self-selected online population that is likely not representative of the general population. Participants were disproportionately well-educated (68% reporting at least some college education) and predominantly from the United States. However, we did not find any substantial differences in age trends in our subgroup analyses. Regarding our modeling approach, we chose to restrict k to positive values, making it a marker of self-excitation. The main reasons for this are methodological: negative k values (indicating leakage of evidence) are harder to recover reliably, and it was necessary to make the OUM clearly differentiable from the DDM (with $k = 0$) for the model comparison. Still, negative k values might be theoretically meaningful and important, for example, in clinical aging populations, and should therefore be explored in future studies.

It must also be noted that k and the boundary separation parameter a were difficult to dissociate in simulated data, as indicated by large correlations of the within-person joint posterior distributions of the parameters. This might partly be a weakness of our chosen data sets, which both contained relatively low numbers of experimental trials for person and task. Importantly, across-person correlations of boundary separation and self-excitation were low, indicating that the recoverability issues might be largely orthogonal to individual differences (and thus age differences) in these parameters. Future studies might use a) other data with higher trial numbers and/or b) a Bayesian hierarchical cognitive model utilizing data from several participants at the same time to obtain more reliable estimates due to shrinkage. While such an approach would have been feasible in our 125 participant data set, it is with current software close to impossible for 5.5 million participants (although see Arruda et al. (2025) for a promising method).

Another constraint concerns the restriction of our findings to binary choice tasks. While this is — due to the model setup — standard practice in age studies utilizing the DDM, k might manifest differently in multi-alternative or continuous-entry decisions. Finally, and probably most importantly, our analyses are, of course, based only on behavioral data. To further explore the suggested theoretical links between the behavioral

and the neurophysiological levels, and especially with regard to the possible link between dopamine and k, studies directly correlating PET/fMRI, biomarkers of dopaminergic and noradrenergic function, and behavioral markers are needed. SBI might be particularly suited for jointly modeling RT and neurophysiological data in an integrative, multi-level approach, through its inherent ability to utilize in principle arbitrarily complex models.

To conclude, our studies suggest that reduced self-excitation may constitute an important, previously overlooked component of cognitive aging in evidence accumulation modeling, alongside increased decision caution and non-decision slowing. Older participants seem to lose some of the "rushed" efficiency brought about by the higher self-excitation of youth, at the same time winning a more objective, unbiased lens. The systematic decline in k may offer a computational bridge between behavioral RT patterns and the neurobiological degradation of neural gain control. Potentially, k might even serve as a digital biomarker for the health of neuromodulatory systems and early neurodegenerative decline. We hope that these findings encourage researchers to further explore the OUM and its implications for understanding the dynamics of cognitive aging.

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Appendix A

Study 1: Model Comparison Validation and Parameter Recovery Validation

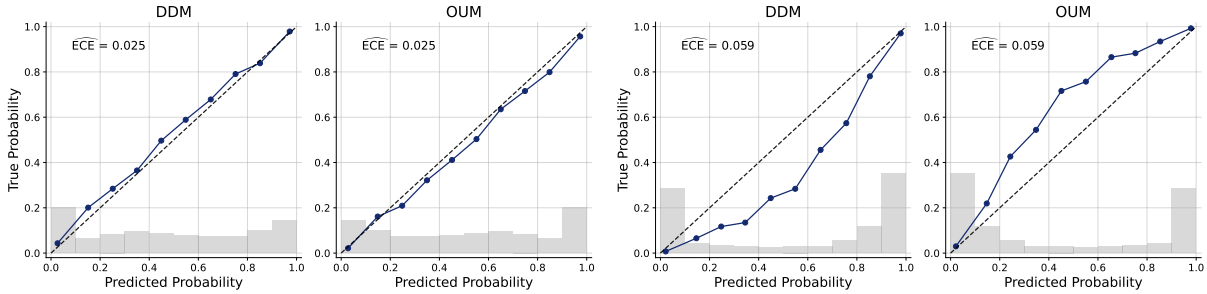

Figure A1

Calibration of the Model Comparison Classifier for Study 1

Note. Calibration plots for the neural network classifier trained to distinguish between the DDM and OUM. Left: Fast tasks. Right: Slow tasks. The plots show the relationship between predicted posterior model probabilities and true model identity in 500 simulated validation data sets. Well-calibrated classifiers fall along the diagonal.

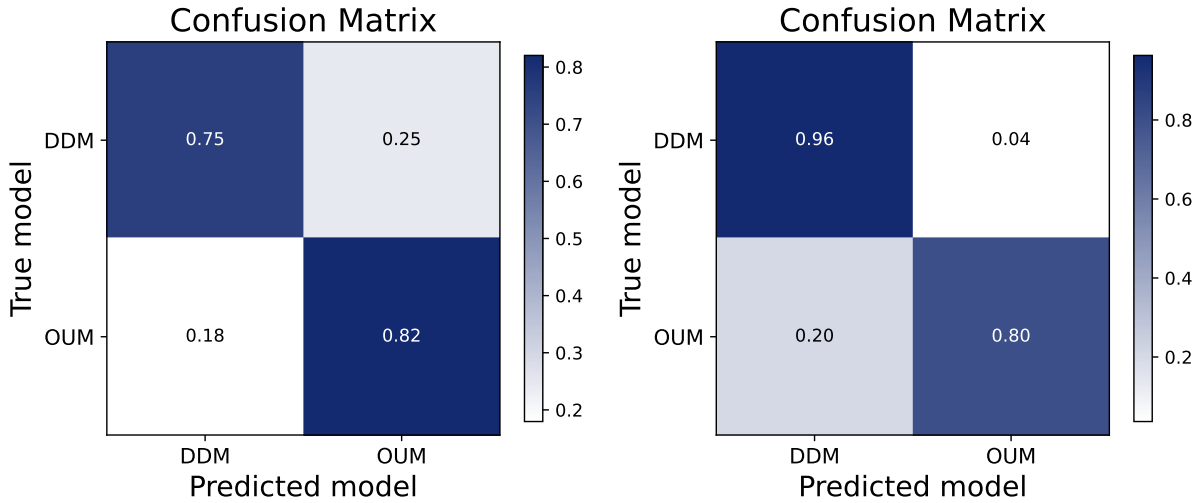

Figure A2

Confusion Matrices for the Model Comparison Classifier in Study 1

Note. Confusion matrices showing classification accuracy for the DDM and OUM in 500 simulated validation data sets. Left: Fast tasks. Right: Slow tasks. The matrices indicate good discernibility of the two cognitive models.

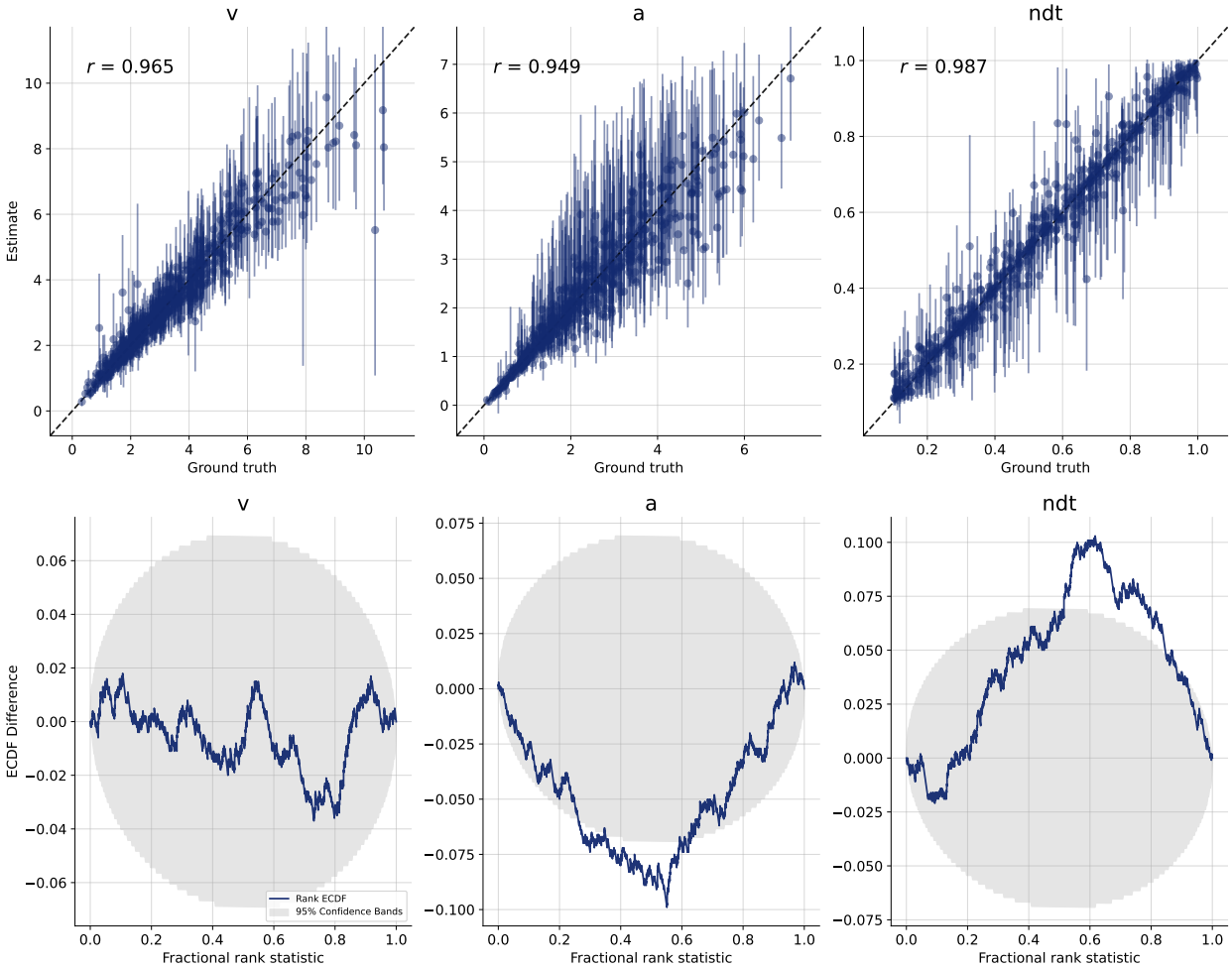

Figure A3

Parameter Recovery and Calibration for the DDM in Fast Tasks

Note. Top: Scatter plots of true vs. estimated parameter values for the DDM in 1000 simulated fast-task data sets. Down: Simulation-based calibration (ECDF) plots for the DDM posteriors. Well-calibrated posteriors produce ECDF distributions within the gray area.

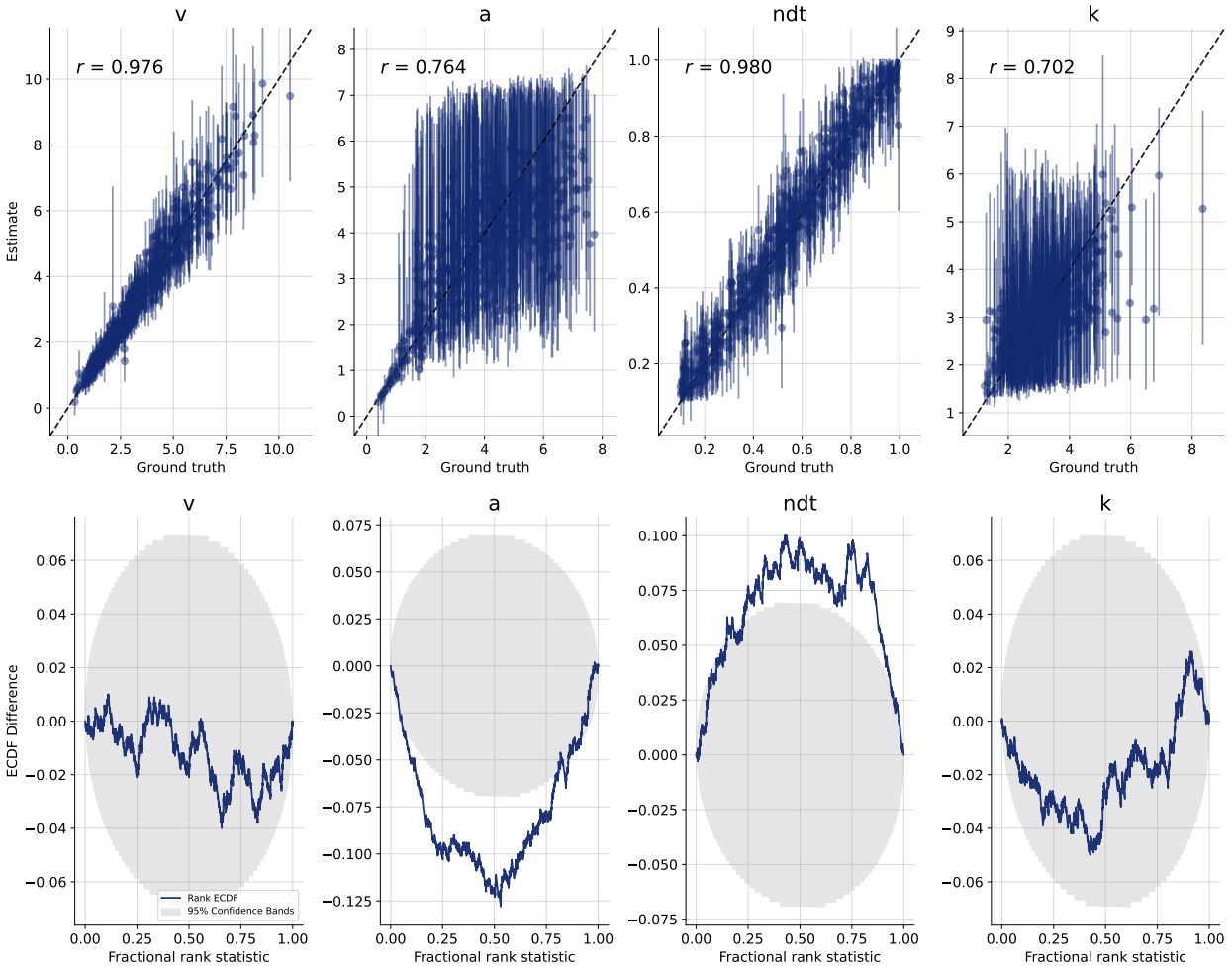

Figure A4

Parameter Recovery and Calibration for the OUM in Fast Tasks

Note. Top: Scatter plots of true vs. estimated parameter values for the OUM in 1000 simulated fast-task data sets. Down: Simulation-based calibration (ECDF) plots for the OUM posteriors. Recovery for boundary separation is slightly worse in the OUM compared to the DDM.

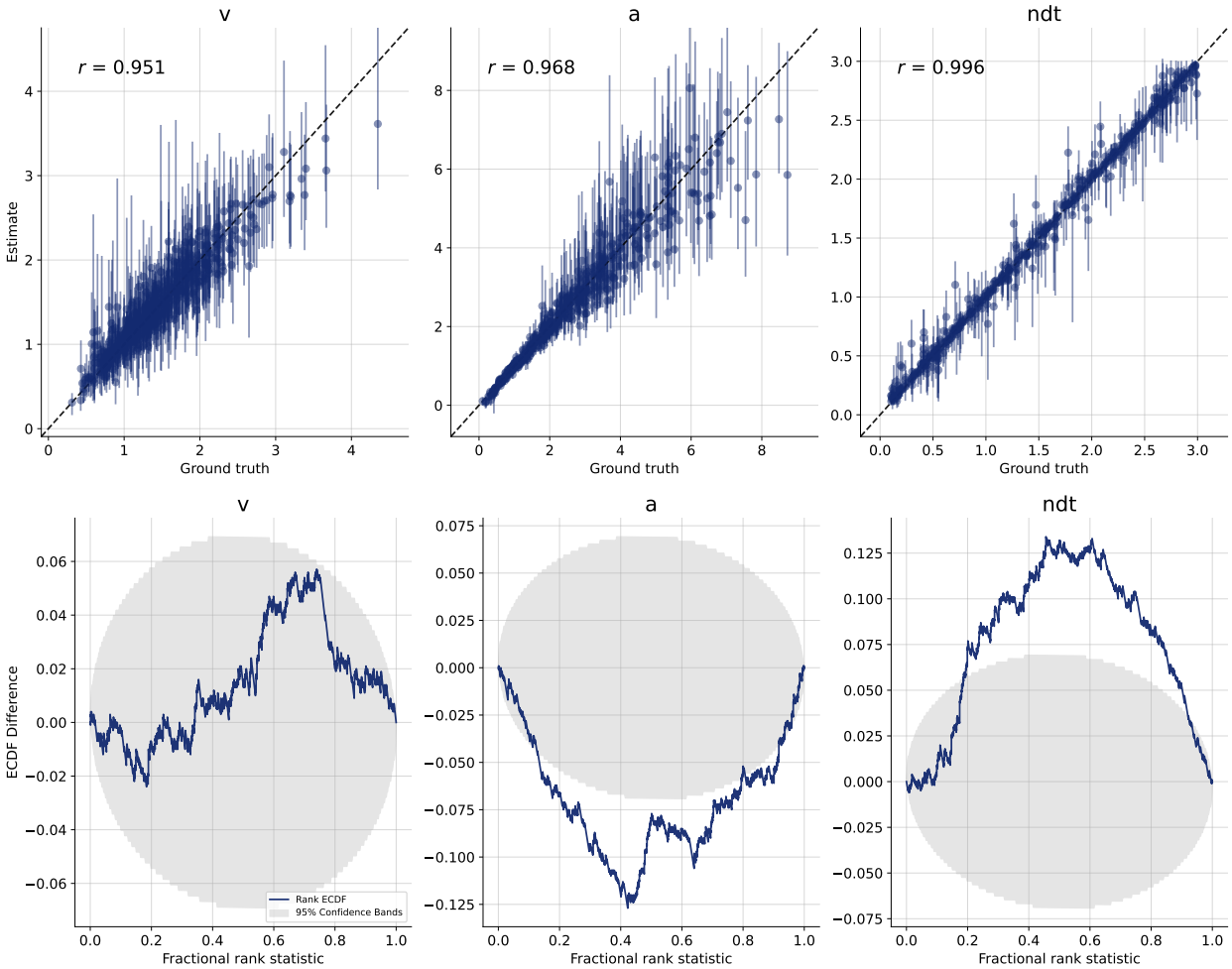

Figure A5

Parameter Recovery and Calibration for the DDM in Slow Tasks

Note. Top: Scatter plots of true vs. estimated parameter values for the DDM in 1000 simulated slow-task data sets. Down: Simulation-based calibration (ECDF) plots for the DDM posteriors.

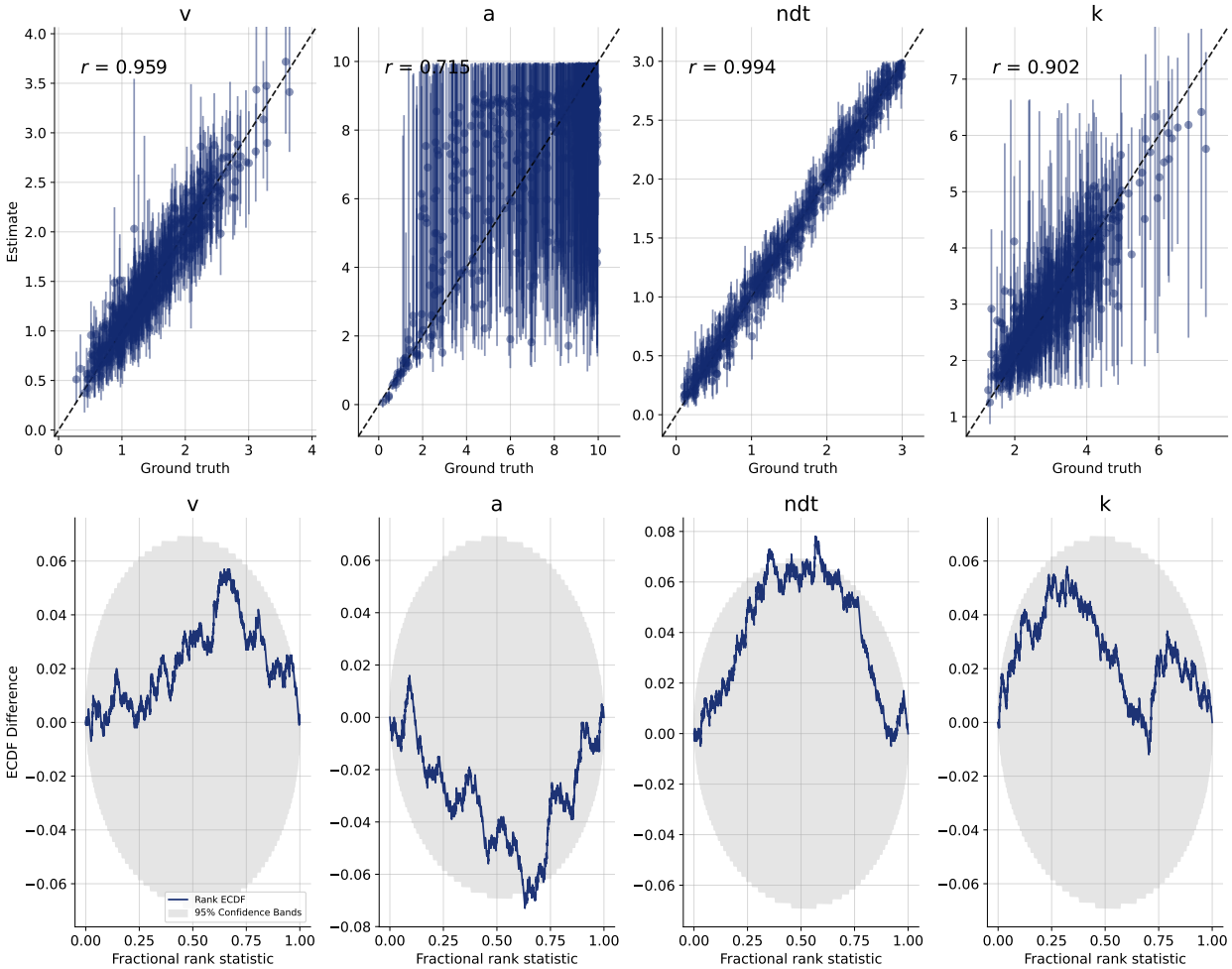

Figure A6

Parameter Recovery and Calibration for the OUM in Slow Tasks

Note. Top: Scatter plots of true vs. estimated parameter values for the OUM in 1000 simulated slow-task data sets. Down: Simulation-based calibration (ECDF) plots for the OUM posteriors. As with fast tasks, boundary separation recovery is worse in the OUM.

Appendix B

Study 1: Posterior Predictive Checks

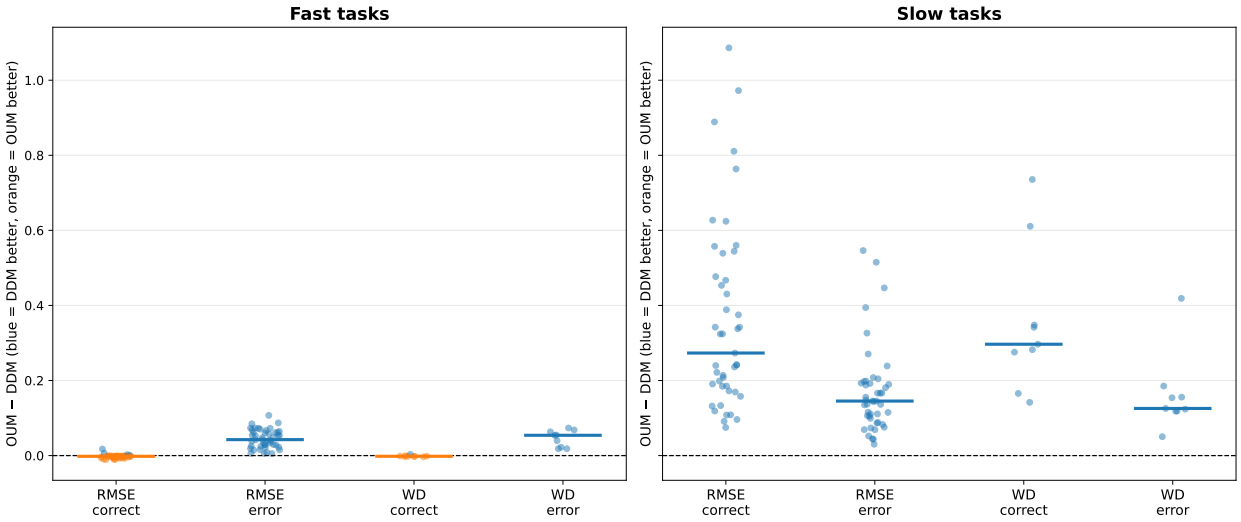

Figure B1

Posterior Predictive Check Comparison for Study 1

Note. DDM − OUM differences in posterior predictive check metrics across all 18 RT tasks. Positive values indicate better DDM fit; negative values indicate better OUM fit. Red lines show medians; blue dashed lines show means. WD = Wasserstein distance for correct or error RTs. RMSE = Mean root mean squared error across the 10th, 30th, 50th, 70th, and 90th percentiles of correct or error RTs.

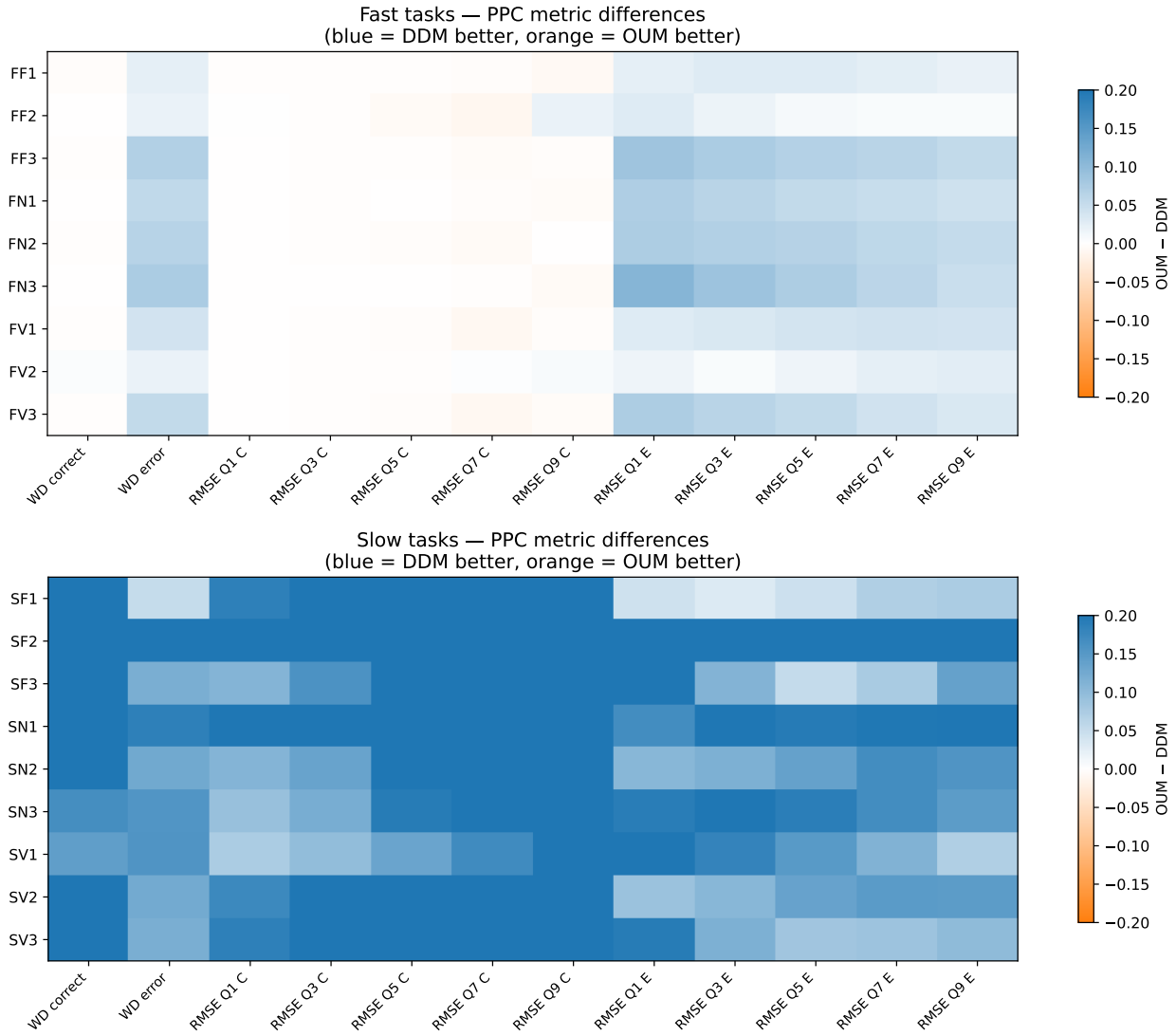

Figure B2

Posterior Predictive Check Heatmaps for Study 1

Note. Heatmaps of DDM − OUM differences in PPC metrics for each individual task. Top: Fast tasks. Down: Slow tasks. Positive values (warmer colors) indicate better DDM fit; negative values (cooler colors) indicate better OUM fit. WD = Wasserstein distance for correct or error RTs. RMSE = Root mean squared error for 10th, 30th, 50th, 70th, and 90th percentile of correct or error RTs.

Appendix C

Study 1: Age Correlations Table

Table C1

Median posterior age correlations [95% CI] for DDM and OUM parameters across 18 RT tasks (Study 1).

Panel A: DDM

Task	ν	a	τ
Fast tasks			
FF1	−0.11 [−0.18, −0.03]	+0.48 [+0.41, +0.53]	+0.56 [+0.51, +0.60]
FF2	−0.31 [−0.36, −0.25]	+0.35 [+0.26, +0.43]	+0.45 [+0.39, +0.50]
FF3	−0.06 [−0.14, +0.01]	+0.45 [+0.37, +0.51]	+0.27 [+0.18, +0.35]
FN1	+0.01 [−0.06, +0.07]	+0.29 [+0.21, +0.36]	+0.36 [+0.28, +0.44]
FN2	+0.10 [+0.03, +0.16]	+0.17 [+0.08, +0.26]	+0.38 [+0.31, +0.44]
FN3	+0.08 [+0.02, +0.14]	+0.39 [+0.31, +0.46]	+0.46 [+0.38, +0.52]
FV1	+0.14 [+0.08, +0.20]	+0.40 [+0.33, +0.46]	+0.31 [+0.23, +0.38]
FV2	+0.06 [+0.00, +0.11]	+0.42 [+0.35, +0.48]	+0.14 [+0.08, +0.19]
FV3	−0.06 [−0.12, +0.01]	+0.38 [+0.27, +0.47]	+0.35 [+0.28, +0.42]
Slow tasks			
SF1	−0.28 [−0.35, −0.20]	+0.45 [+0.39, +0.51]	+0.25 [+0.17, +0.33]
SF2	−0.07 [−0.14, +0.00]	+0.34 [+0.30, +0.39]	+0.21 [+0.14, +0.28]
SF3	−0.03 [−0.11, +0.04]	+0.31 [+0.27, +0.35]	+0.20 [+0.12, +0.28]
SN1	−0.06 [−0.12, +0.01]	+0.16 [+0.10, +0.21]	+0.17 [+0.09, +0.24]
SN2	+0.01 [−0.05, +0.07]	+0.26 [+0.21, +0.31]	+0.33 [+0.26, +0.41]
SN3	−0.03 [−0.09, +0.03]	+0.21 [+0.15, +0.28]	+0.49 [+0.43, +0.54]
SV1	+0.08 [−0.01, +0.17]	+0.25 [+0.18, +0.32]	+0.27 [+0.21, +0.32]
SV2	−0.29 [−0.35, −0.22]	+0.42 [+0.36, +0.47]	+0.48 [+0.37, +0.55]
SV3	−0.07 [−0.14, −0.00]	+0.34 [+0.28, +0.41]	+0.39 [+0.31, +0.46]

Panel B: OUM

Task	ν	a	τ	k
Fast tasks				
FF1	−0.14 [−0.20, −0.06]	+0.38 [+0.27, +0.47]	+0.25 [+0.14, +0.35]	−0.08 [−0.21, +0.05]
FF2	−0.34 [−0.39, −0.28]	+0.36 [+0.24, +0.47]	+0.18 [+0.05, +0.30]	−0.26 [−0.36, −0.15]
FF3	−0.16 [−0.23, −0.08]	+0.43 [+0.33, +0.52]	+0.01 [−0.11, +0.13]	−0.17 [−0.29, −0.04]
FN1	−0.03 [−0.09, +0.04]	+0.33 [+0.24, +0.43]	+0.12 [+0.01, +0.22]	−0.14 [−0.28, +0.01]
FN2	+0.11 [+0.04, +0.17]	+0.17 [+0.06, +0.28]	+0.21 [+0.10, +0.31]	−0.11 [−0.25, +0.03]
FN3	+0.04 [−0.02, +0.09]	+0.43 [+0.33, +0.51]	+0.25 [+0.14, +0.35]	−0.21 [−0.34, −0.08]
FV1	+0.14 [+0.08, +0.20]	+0.34 [+0.23, +0.44]	+0.10 [−0.03, +0.21]	−0.18 [−0.29, −0.07]
FV2	+0.06 [+0.00, +0.12]	+0.29 [+0.18, +0.39]	+0.01 [−0.11, +0.13]	−0.35 [−0.44, −0.26]
FV3	−0.08 [−0.14, −0.01]	+0.32 [+0.20, +0.43]	+0.06 [−0.06, +0.18]	−0.22 [−0.33, −0.11]
Slow tasks				
SF1	−0.26 [−0.37, −0.15]	−0.10 [−0.25, +0.04]	+0.24 [+0.10, +0.37]	−0.34 [−0.46, −0.20]
SF2	−0.03 [−0.16, +0.10]	−0.05 [−0.18, +0.06]	+0.17 [+0.01, +0.32]	−0.29 [−0.38, −0.19]
SF3	+0.05 [−0.05, +0.16]	+0.13 [+0.01, +0.26]	+0.15 [−0.01, +0.30]	−0.24 [−0.33, −0.16]
SN1	−0.00 [−0.12, +0.12]	−0.04 [−0.16, +0.08]	+0.10 [−0.03, +0.24]	−0.17 [−0.27, −0.07]
SN2	+0.01 [−0.08, +0.10]	+0.10 [−0.05, +0.25]	+0.18 [+0.02, +0.33]	−0.19 [−0.29, −0.08]
SN3	−0.00 [−0.07, +0.08]	+0.08 [−0.08, +0.22]	+0.27 [+0.14, +0.40]	−0.16 [−0.25, −0.07]
SV1	+0.01 [−0.09, +0.11]	+0.11 [−0.04, +0.26]	+0.22 [+0.09, +0.34]	−0.21 [−0.33, −0.09]
SV2	−0.30 [−0.38, −0.21]	−0.06 [−0.22, +0.08]	+0.23 [+0.07, +0.38]	−0.37 [−0.48, −0.27]
SV3	−0.07 [−0.17, +0.03]	−0.01 [−0.16, +0.14]	+0.20 [+0.03, +0.35]	−0.27 [−0.39, −0.14]

Note. Each cell shows the median posterior Pearson correlation with the 95% credible interval in brackets. ν = drift rate, a = boundary separation, τ = non-decision time, k = self-excitation (OUM only).

Appendix D

Study 2: Model Comparison and Parameter Recovery Validation

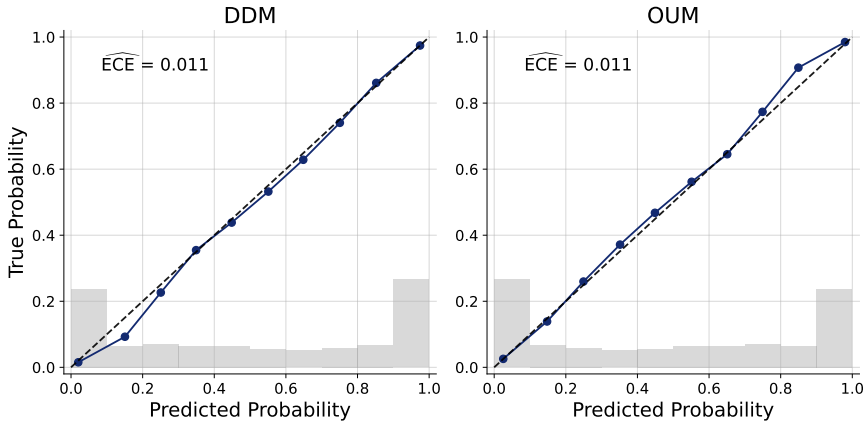

Figure D1

Calibration of the Model Comparison Classifier for Study 2 (IAT)

Note. Calibration plot for the neural network classifier trained to distinguish between the DDM and OUM in the IAT data. The plot shows the relationship between predicted posterior model probabilities and true model identity in 10,000 simulated validation data sets. Well-calibrated classifiers fall along the diagonal.

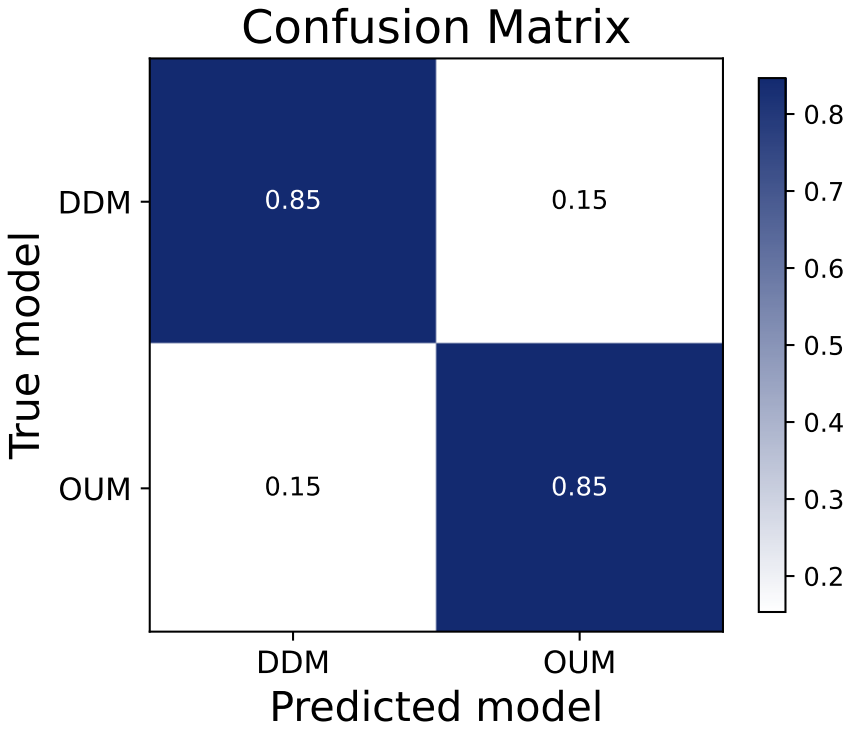

Figure D2

Confusion Matrix for the Model Comparison Classifier in Study 2 (IAT)

Note. Confusion matrix showing classification accuracy for the DDM and OUM in 10,000 simulated validation data sets. The matrix indicates the classifier can distinguish between the two models, though not perfectly.

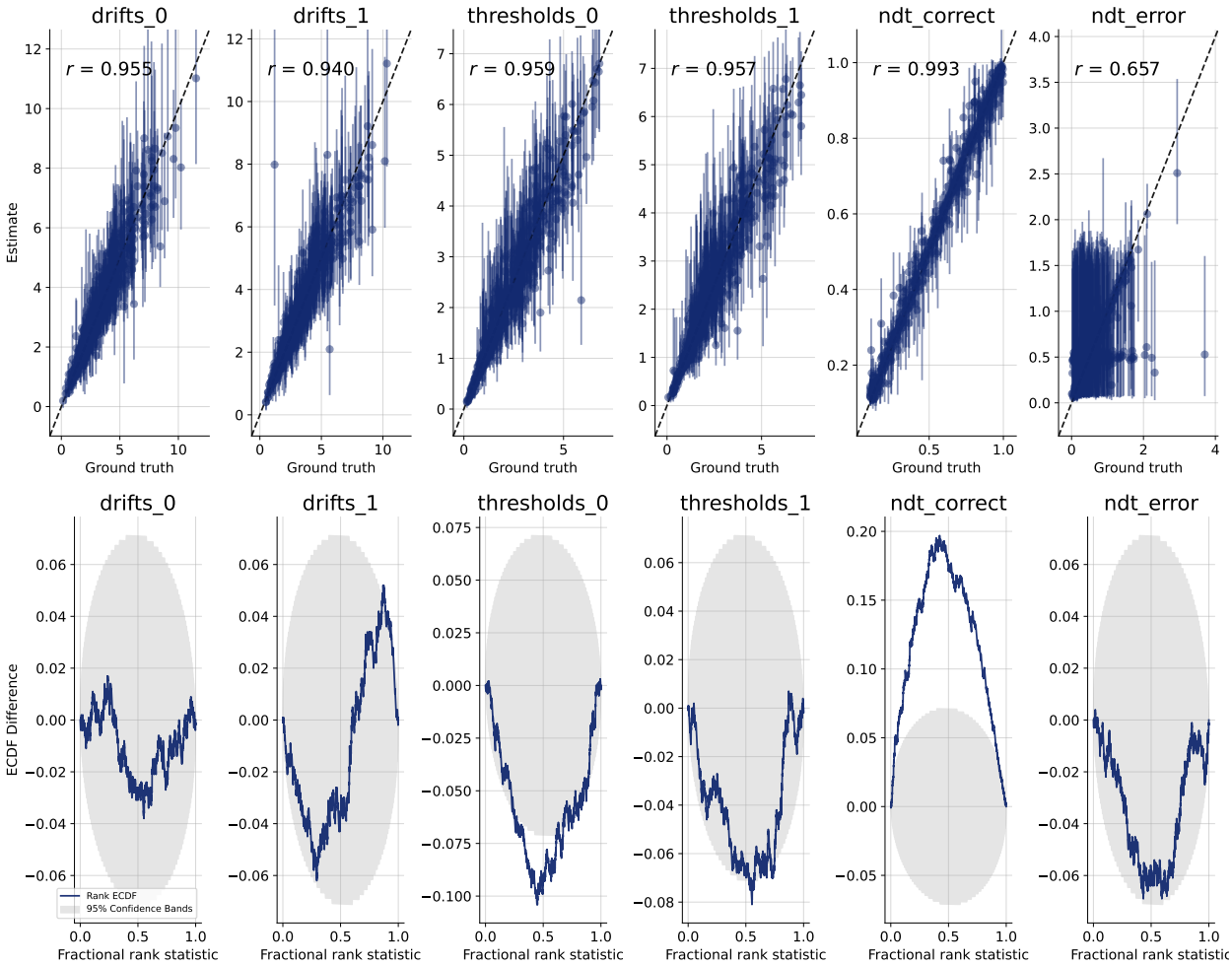

Figure D3

Parameter Recovery and Calibration for the DDM in Study 2 (IAT)

Note. Top: Scatter plots of true vs. estimated parameter values for the DDM in 500 simulated IAT data sets. Down: Simulation-based calibration (ECDF) plots for the DDM posteriors.

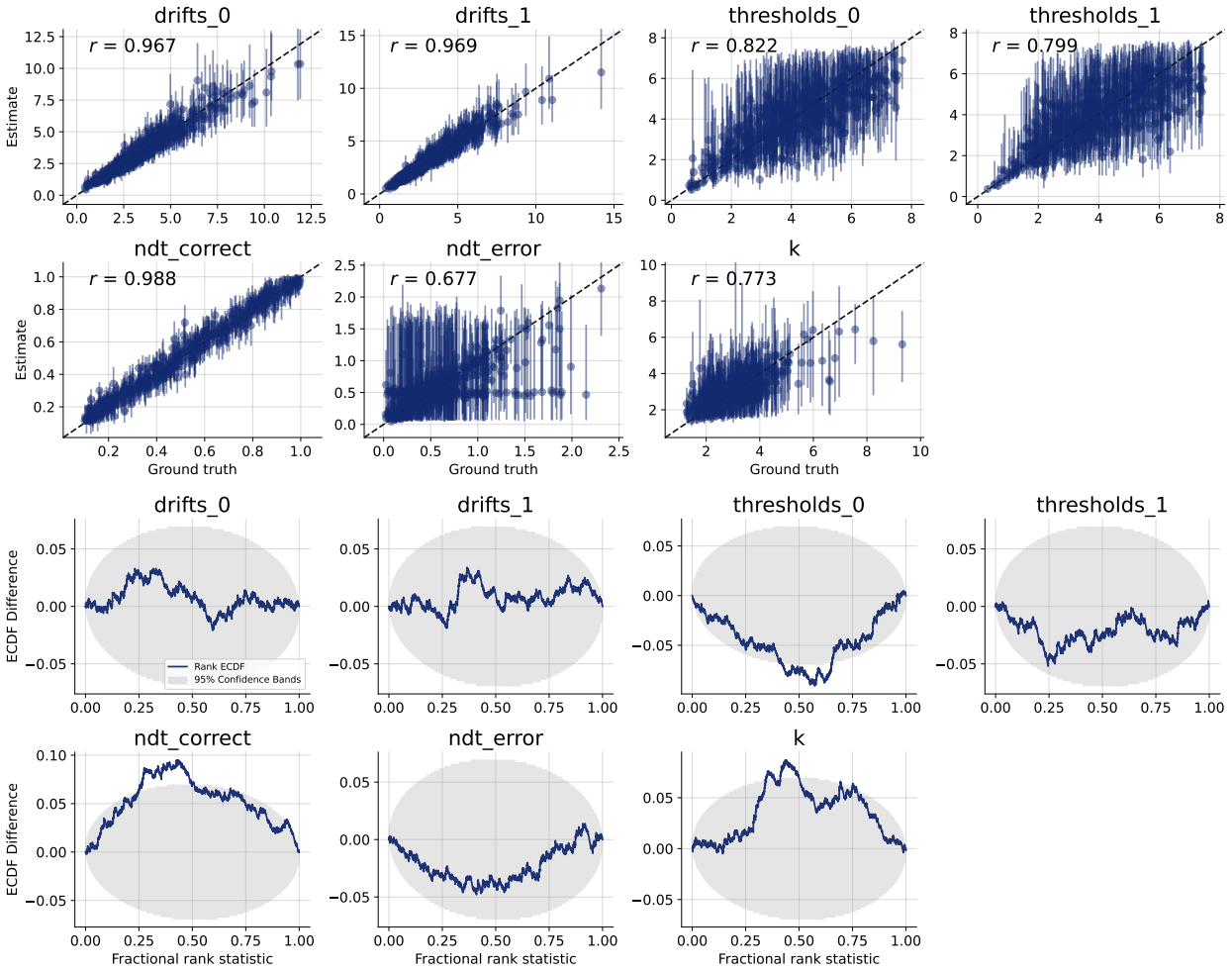

Figure D4

Parameter Recovery and Calibration for the OUM in Study 2 (IAT)

Note. Top: Scatter plots of true vs. estimated parameter values for the OUM in 500 simulated IAT data sets. Down: Simulation-based calibration (ECDF) plots for the OUM posteriors. Recovery for boundary separation is slightly worse than for other parameters.

Appendix E

Study 2: Posterior Predictive Checks

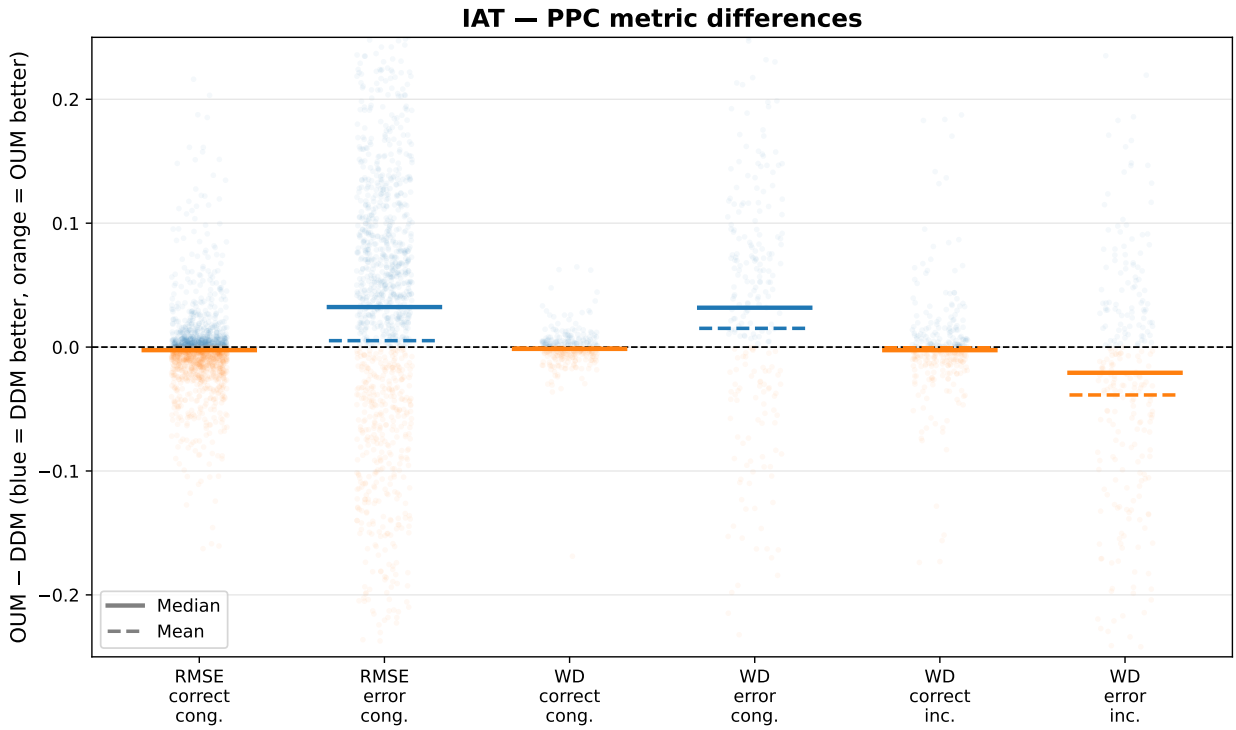

Figure E1

Posterior Predictive Check Comparison for Study 2 (IAT)

Note. DDM − OUM differences in posterior predictive check metrics computed on a random subsample of 100 participants per data chunk (100 PPC simulations each). Positive values indicate better DDM fit; negative values indicate better OUM fit. Red lines show medians; blue dashed lines show means. WD = Wasserstein distance for correct or error RTs. RMSE = Mean root mean squared error across the 10th, 30th, 50th, 70th, and 90th percentiles of correct or error RTs in congruent or incongruent trials.